Chapter 2

Applied basic science

Surface anatomy *34*
Thoracic bony cage *36*
Thoracic incisions *38*
Mediastinum *40*
Cardiac chambers *42*
Cardiac valves *44*
Coronary arteries and veins *46*
Lung anatomy *48*
Anatomy of the esophagus *50*
Thoracic vessels *52*
Vascular anatomy and pathology *54*
Myocardium *56*
The electrocardiogram *58*
Cardiac function *60*
Circulation *62*
Regulation of blood pressure *64*
Lung and esophageal physiology *66*
Renal homeostasis *68*
Blood *70*
Ischemic heart disease *72*
Ischemia–reperfusion injury *74*
Pathology of heart failure *76*
Valvular heart disease *78*
Infective endocarditis *80*
Pericardial pathology *82*
Cardiac neoplasms *84*

Anatomy of the chest wall, airways, lungs and pleura, and esophagus are described in more detail on 📖 pp655–661.
Embryology is described on 📖 pp528–535.

Surface anatomy

Bony landmarks

- The 1st rib is difficult to palpate as it lies under the clavicle. The 2nd rib articulates with the manubrium, just above the sternal angle. The 1st rib space is palpable just above and lateral to the sternal angle. The costal margin is formed by lower borders of ribs 7–10, and ends of 11 and 12. Important bony landmarks include cricoid cartilage (C6), suprasternal notch (T2), sternal angle (T4/5), xiphisternal joint (T9) anteriorly, and the 1st palpable spinous process (vertebra prominens), C7 (C1–6 are covered by the ligamentum nuchae), and the superior (T2) and inferior angle (T8) and spine (T3) of the scapula posteriorly.

Lines of orientation

The midclavicular line is a vertical line from the midpoint of the clavicle downwards. The anterior and posterior axillary lines are from anterior and posterior axillary folds downwards. The midaxillary line lies midway between the anterior and posterior axillary lines.

Surface markings of thoracic structures

Trachea

This commences at the lower border of the cricoid cartilage (C6), descending in the midline to end slightly to the right, bifurcating at the level of the sternal angle (T4/5) into left and right bronchi.

Lungs and pleurae

The apex of the pleura curves 2.5cm above the medial 1/3 of the clavicle. Lines of pleural reflection pass behind sternoclavicular joints meeting in the midline at the sternal angle. The right pleura passes down behind the 6th costal cartilage whereas the left, displaced by the heart, passes laterally for 2cm at the 4th costal cartilage then descending to the 6th costal cartilage lateral to the left sternal edge. From here both pleura run posteriorly crossing the 8th rib in the midclavicular line, the 10th rib in the midaxillary line and just below the 12th rib at the medial border of the erector spinae muscle posteriorly above the pleural reflections. This varies by 5–8cm in extremes of respiration. Both oblique fissures run from T3 posteriorly to the 6th costal cartilage in the midaxillary line. The horizontal fissure roughly follows the 5th rib.

Heart

The heart is bounded by the 2nd left costal cartilage, the 3rd right costal cartilage, the 6th right costal cartilage, and the 5th left costal cartilage.

Vessels

The internal thoracic arteries descend behind the costal cartilages, 1cm lateral to the sternal edge. The aortic arch arches anteroposteriorly behind the manubrium, the innominate, and left common carotid ascend posterior to the manubrium. The innominate veins are formed by the confluence of the internal jugular and subclavian veins posterior to the sternoclavicular joints. The SVC arises from the left and right innominate veins behind the 2nd and 3rd right costal cartilages.

Diaphragm

The highest part of the right hemidiaphragm reaches the upper border of the 5th rib in the midclavicular line in mid-inspiration. The left dome reaches the lower border of the 5th rib (see Fig. 2.1).

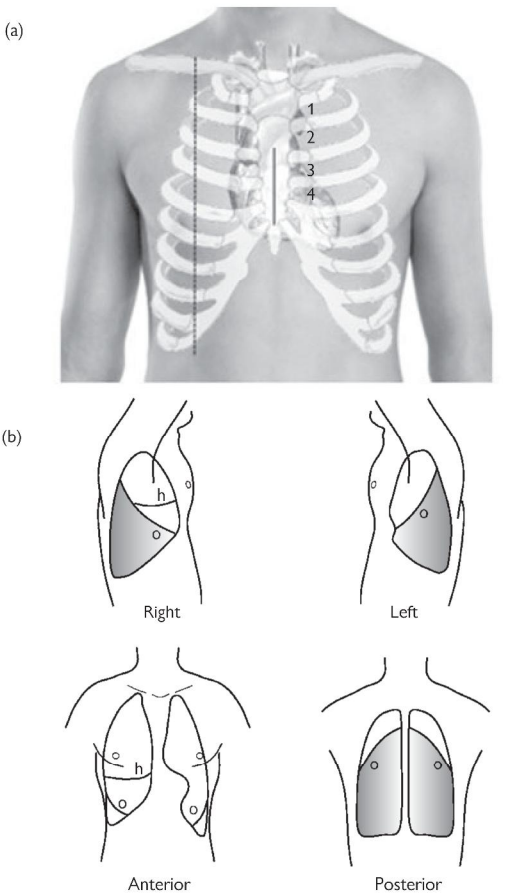


Fig. 2.1 (a) Surface markings of the heart and great vessels. Dotted line represents midaxillary line. Solid line represents standard low midline incision for full sternotomy. Numbers indicate intercostal spaces. (b) Surface markings of the lungs (shaded area represents lower lobe).

Thoracic bony cage

The thoracic cage

This is formed by the sternum and costal cartilages anteriorly, the vertebral column posteriorly, and the ribs and intercostal spaces laterally. It is separated from the abdominal cavity by the diaphragm and communicates superiorly with the root of the neck via the ***thoracic inlet***.

Ribs

There are 12 pairs of ribs:

- 7 pairs of true ribs which articulate with the sternum via costal cartilages and the vertebrae.
- The false ribs whose cartilage articulates with that of the rib above.
- The false floating ribs 11 and 12.

Typical ribs

Typical ribs comprise:

- A head bearing two articular facets which articulate with the corresponding vertebrae and the one above.
- A neck which gives attachment to the costotransverse ligaments.
- A tubercle which has a smooth facet for articulation with the transverse process of the corresponding vertebrae.
- A long shaft with a costal groove in which the intercostal vessels and nerve runs, just *below* the body of the rib.

The atypical ribs

- The atypical ribs are 1, 2, 10, 11, and 12. The 1st rib is short and flat. The head has a single facet, the scalenus anterior inserts onto the prominent scalene tubercle. The subclavian vein passes over the 1st rib anterior to this, the subclavian artery and lowest trunk of the brachial plexus pass posteriorly (Fig. 2.2). A *cervical rib* is an extra rib which articulates with C7 posteriorly and the 1st rib anteriorly.

Intercostal nerves and arteries

The intercostal nerves are the anterior primary rami of the thoracic nerves. They give off a collateral muscular branch and lateral and anterior cutaneous branches (see Table 2.1). Posterior intercostal arteries are branches of the thoracic aorta, except for the first two which are arise from the costocervical trunk. The six anterior intercostals are branches of the internal mammary artery (IMA).

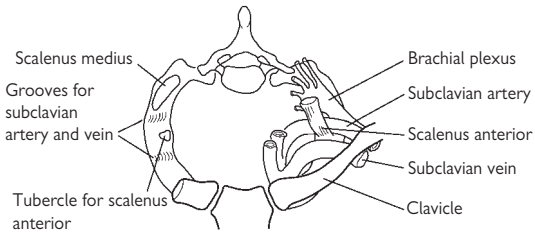


Fig. 2.2 Structures crossing the first rib.

Table 2.1 The muscles of the thoracic wall, innervation, and attachments

Muscle	Origin/insertion	Nerve (n.)/ artery (a.)	Action
Latissimus dorsi	Vertebral spines T7–sacrum, iliac crest, lower 4 ribs. Intertubercular groove	Thoracodorsal n. and a.	Extends arm, rotates medially
Serratus anterior	Ribs 1–9, medial border of scapula on costal surface	Long thoracic nerve of Bell (C5–7)	Draws scapula forward and rotates
Trapezius	Superior nuchal line, occipital protuberance, spinous processes C7–T12 Lateral 1/3 clavicle, scapular spine	Transverse cervical a. Spinal accessory n.	Elevates, depresses, retracts and rotates scapular superiorly
Diaphragm	Xiphoid, costal margin, lateral and medial arcuate ligaments, vertebral bodies 1–3/central tendon of diaphragm	C3–5 phrenic n. and a.	Descends to produce inspiration
Pectoralis major	Clavicle, manubrium and sternum, costal cartilages 2–6, rectus sheath. Crest of great tubercle of humerus	Pectoral n: C5–T1, pectoral a., thoracoacromial a.	Flexes, adducts, and medially rotates arm
External intercostals	Lower border of rib above/downwards and medially to upper border of rib below	Intercostals n. (T1–11) and a	Elevates ribcage and strengthens wall
Internal intercostals	Upper border of rib below/upwards and laterally to lower border of rib above	Intercostals n. (T1–11) and a.	Elevates and reinforces ribcage
Innermost intercostals	Same as internal intercostals but deep to the intercostal a. and n.	Intercostals n.(T1–11) and artery	Elevates and reinforces ribcage
Levators costarum	Transverse processes C7–11/rib below, medial to angle	Dorsal rami C7–11	Elevates the rib
Levator scapulae	C1–4 transverse processes, medial border scapula	Dorsal scapular n. and a.	Elevates scapula
Rhomboids min and maj	Ligamentum nuchae, spines C7–T1, T2–5 Medial border of scapula	Dorsal scapular n. and a.	Retracts, elevates and rotates the scapula inferiorly
Erector spinae	Iliac crest, sacrum, transverse, and spinous processes, supraspinal ligament, rib angles	Segmental C1–S5	Extends and laterally bends spine

Thoracic incisions

The chest may be entered via anterior, lateral, or posterior approaches.

Anterior incisions

- ***Median sternotomy:*** vertical midline incision from suprasternal notch to lower end of xiphoid. The sternum is divided using a sternal saw.
  - Identify midline: drape symmetrically so you can see landmarks, and palpate sternal notch, 2nd intercostal spaces, and xiphisternum
  - 23 blade to divide epidermis barely breaking fat (generally an incision from 3 fingerbreadths below sternal notch to the tip of xiphisternum is more than adequate for most cardiac surgery).
  - Cautery on 50W (fulgurate). Spread tissue with left hand and cauterize down to periosteum.
  - Repalpate midline landmarks and carefully incise midline of periosteum with cautery top to bottom: this marks a line for the saw.
  - Sweep pericardiacosternal ligaments from under xiphisternum with a finger. Insert sternal saw under xiphisternum, and lifting slightly upwards push it forwards to divide sternum. If a low incision you will need an assistant to retract skin at top. The saw can be used top to bottom, in which case there is no need to sweep under xiphisternum—instead, clear sternal notch of tissue.
  - The sternal saw is designed to stop if caught in soft tissue: so do not force it as you may end up opening the pericardium or worse.
  - Place pack under bone edges, cauterize periosteum, and use bone wax judiciously to stop bleeding marrow.
  - Re-sternotomy is described on 📖 p345.
- ***Transverse cervical or collar incision:*** transverse incision midway between suprasternal notch and thyroid cartilage, along skin crease. This incision is used for thyroidectomy, tracheostomy (smaller and more inferior), and upper half tracheal resection and reconstruction.
- ***Left anterior to sternocleidomastoid incision:*** oblique incision anterior to the sternocleidomastoid incision, extending to the sternal notch. This incision is used for exposure of the cervical esophagus, or anterior cervical spine.
- ***Dartevelle:*** longitudinal incision along anterior border of sternocleidomastoid, continuing transversely across to the medial clavicle. The clavicle can be mobilized and distracted from the incision.
- ***Anterior mediastinotomy:*** transverse incision along 2nd intercostal space laterally from parasternal edge. Used for a Chamberlain mediastinal biopsy. Often the sternal–rib cartilage junction is removed. Care is taken to preserve the internal mammary artery.
- ***Anterior thoracotomy:*** transverse incision from parasternal edge along the inframammary crease to the anterior axillary line in the 4th interspace. The pectoralis major muscle can be elevated off the chest.
- ***Thoracosternotomy ('clamshell'):*** incision from anterior axillary line along submammary crease bilaterally elevating in the midline to the level of the nipple (📖 p819). As an anterior thoracotomy, the pectoralis major can be elevated and raised a flap. The sternum is divided transversely, and reconstructed with wires ± pins.

Lateral and posterior incisions

- ***Axillary thoracotomy:*** curvilinear incision at the inferior border of axillary hairline between lateral border of pectoralis major and anterior border of latissimus dorsi.
- ***Lateral 'muscle-sparing' thoracotomy: lazy 'S':*** incision along submammary crease starting below nipple passing upwards towards the axilla.
- ***French:*** incision along submammary crease from below nipple to 2cm below tip of scapula. The muscle-sparing thoracotomy is characterized by mobilization and preservation of the latissimus dorsi muscle. The serratus anterior muscle is divided in the direction of its fibres.
- ***Lateral thoracotomy:*** curvilinear incision passing from anterior axillary line to 2cm below tip of scapula.
- ***Posterolateral thoracotomy:*** curvilinear incision from anterior axillary line passing 3–4cm below tip of scapula and continuing superiorly midway between medial border of scapula and spinous processes of vertebral column. This standard 'workhorse' incision usually involves division of the latissimus dorsi muscle, and distal division (or sparing) of the serratus anterior muscle.
- ***Thoracoabdominal incision:*** oblique incision as for posterolateral incision but extending anteriorly across the costal margin at the level of the 7th interspace towards the midline.

Operative positions

- ***Lateral decubitus position*** (side up): most common position used in thoracic surgery. Allows the best surgical access to and control of structures in the hilum. Ventilation of the contralateral dependent lung may be more troublesome than with other positions. The ipsilateral arm is extended, and rested on a cushioned platform or multiple folded blankets. Care must be taken to avoid stretch of the ipsilateral brachial plexus from overextension, as well as crush of the down brachial plexus. This may be avoided by placement of a transverse shoulder roll, just inferior to the down axilla. The bed is flexed to spread the intercostal space, and drop the ipsilateral hip out of the way. All pressure points are padded, and in males the testicles are distracted anteriorly.
- ***Prone*** (face down): allows access to the posterior structures, specifically the spine and esophagus. Major vessels are furthest away from operator and access to the hilum is limited. Care is taken to avoid ocular pressure.
- ***Supine*** (face up): sternotomy easier with roll under shoulders.
- ***Fowler's:*** patient on back, sitting up at about 45–60°, usually with arms extended on boards. Allows video-assisted thoracic surgery (VATS) approach to both pleura, and head-up tilt helps lung fall away from apex to allow apical procedures such as sympathectomy (📖 p804).

Mediastinum

Key facts

- The mediastinum is the space between the pleural sacs.
- It is divided by a line drawn horizontally from the sternal angle to the lower border of T4, into ***superior*** (bounded by the thoracic inlet above) and *inferior* ***mediastinum*** (bounded by the diaphragm below).
- The inferior mediastinum is further divided by the pericardial sac into anterior, middle, and posterior (Fig. 2.3a).

Contents of the mediastinum

- ***Superior mediastinum:*** great vessels, trachea, esophagus, phrenic nerve and vagus nerve, thoracic duct.
- ***Anterior mediastinum:*** sternopericardial ligaments, thymus, lymph nodes.
- ***Middle mediastinum:*** pericardial cavity, heart, great vessels, phrenic nerve.
- ***Posterior mediastinum:*** esophagus, descending aorta, azygos veins, thoracic duct, lymph nodes.

Relationships of the heart

- ***Anteriorly:*** sternum, costal cartilages of ribs 3–5, anterior lungs.
- ***Laterally:*** lungs and hila.
- ***Posteriorly:*** esophagus, tracheal bifurcation, main bronchi, descending aorta, and vertebrae T5–T8.
- ***Inferiorly:*** diaphragm and liver.
- ***Superiorly:*** great vessels.

Pericardium and its reflections

The pericardium has three layers: one fibrous and two internal serous (parietal and visceral) separated by pericardial fluid. The heart and proximal great vessels are contained within the conical fibrous pericardium. The apex fuses with the adventitia of the vessels and the base fuses with the central tendon of the diaphragm. The fibrous pericardium is lined inside by the parietal serous pericardium which is reflected around the great vessels becoming continuous with the visceral layer of pericardium. The lines of reflection define the oblique sinus (bounded by the IVC and four pulmonary veins) and the transverse sinus (bounded by the SVC, LA, pulmonary trunk, and aorta) posteriorly (Fig. 2.3).

Vagus nerve

The vagus nerve contains visceral afferent fibres from the heart, lower respiratory tract and gut, and efferent pre-ganglionic parasympathetic motor fibres to the pharynx, larynx, heart, and smooth muscles of the bronchi and gut. It passes vertically down from the jugular foramen to the root of the neck lying posteriorly in the carotid sheath between the internal jugular vein and the internal then common carotid. In the neck it gives pharyngeal, superior laryngeal, and superior and inferior cardiac branches.

On the right side the recurrent laryngeal branch arises as the vagus crosses the subclavian artery. The vagus descends through the superior mediastinum passing behind the hilum of the right lung to the pulmonary plexus and then the esophageal plexus. The left vagus crosses the arch of the aorta giving off the recurrent laryngeal nerve, which passes below the ligamentum arteriosum, behind the arch and then ascends in the groove between trachea and esophagus. The vagus passes behind the left lung hilum to form the pulmonary and esophageal plexuses. The vagi leave the thorax via the esophageal diaphragmatic opening. The subclavian loop carries afferents from the stellate ganglion to the eye and head, running adjacent to the subclavian bilaterally.

Phrenic nerve

The phrenic nerve contains visceral afferents from the heart and diaphragm, as well as motor efferents from the dorsal rami of C3–5. It enters the thoracic inlet on the anterior surface of the anterior scalene muscle, lying just posterior to the IMA. On the right side the phrenic travels down the front of the SVC descending anterior to the hilum before reflecting onto the right diaphragm. The left phrenic nerve follows a similar course.

Thoracic sympathetic plexus

This lies lateral to the posterior mediastinum behind the parietal pleurae, crossing the necks of the first ribs, the heads of the 2nd–10th ribs and the bodies of the 11th and 12th vertebrae entering the diaphragm to continue as the lumbar sympathetic trunk. It distributes sympathetic branches to the skin, post-ganglionic fibres to thoracic viscera, and pre-ganglionic fibres to the celiac and renal ganglions below the diaphragm.

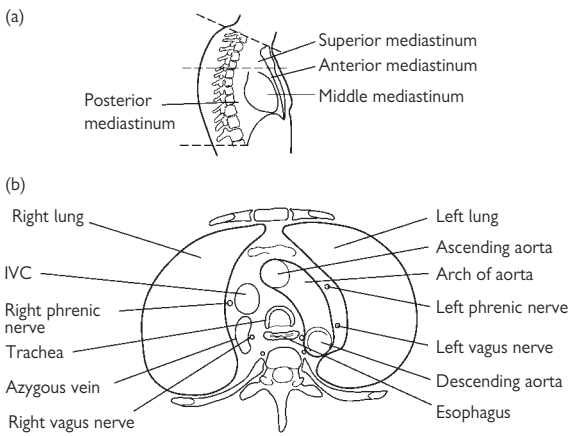


Fig. 2.3 (a) S(ubdivisions of the mediastinum; (b) Cross-section through the mediastinum at T4.

Cardiac chambers

Right atrium

The RA receives venous drainage from four sources:

- The ***SVC*** superiorly, draining the azygos, jugular, and subclavian veins.
- The ***IVC*** inferiorly, draining the lower body.
- The ***coronary sinus*** inferiorly, draining the heart.
- The ***anterior cardiac vein*** anteriorly, draining the front of the heart.

Running vertically downwards between the IVC and SVC is the ***crista terminalis***, a muscular ridge marked externally by the sulcus terminalis. The ***sinus node*** lies at the superior end of the terminal groove, where the atrial appendage meets the SVC. The crista terminalis separates the smooth-walled posterior RA derived from the ***sinus venosus*** from the trabeculated appendage, derived from the fetal RA. The IVC and coronary sinus have rudimentary valves (the ***Eustachian*** and ***Thebesian valves*** respectively). The Eustachian valve is continuous with the annulus ovalis which surrounds the ***fossa ovalis***. It is also continuous with the Thebesian valve via the ***tendon of Todaro***.

The triangle of Koch (Fig. 2.4)

The apex of this long narrow triangle identifies the AV node, and proximal portion of the bundle of His. The triangle is formed by:

- ***Short base:*** coronary sinus and Thebesian valve.
- ***Long upper side:*** septal leaflet of the tricuspid valve.
- ***Long lower side:*** tendon of Todaro.

Right ventricle

The RV consists of a large inlet (sinus) and a smaller outlet (conus) portion separated by the ***supraventricular crest***. The inlet portion surrounds the tricuspid valve and its subvalvular apparatus. The inflow tract is characterized by ***trabeculae carnae***, from some of which ***papillary*** *muscles* project, attaching to the inferior border of the tricuspid valve leaflets via ***tendinae chordae***. The ***moderator band*** is a muscular bundle crossing the ventricular cavity from the interventricular septum to the anterior wall, conveying the right branch of the AV bundle to the ventricular muscle. The outlet portion is smooth walled and consists of the ***infundibulum*** (a muscular structure that supports the ***pulmonary valve***), the superior part of the septal band, and a very narrow portion superior to the trabecular septum. The thick, muscular supraventricular crest separates the pulmonary valve from the tricuspid valve.

Left atrium

The LA is smaller than the right with thicker walls. The four pulmonary veins open into the upper part of its posterior wall. On its septal surface there is a 'flap' valve which corresponds to the fossa ovalis. The ***left atrial appendage*** is characterized by ***pectinate muscles***. It is longer and narrower-based than the wide-based right atrial appendage, so in AF blood has a greater tendency to stagnate inside and form thrombi.

Left ventricle

The LV consists of a large trabeculated inlet (sinus) and a smaller smooth-walled outlet portion. In comparison to the well-separated pulmonary and tricuspid valve, the aortic and mitral valve are in ***fibrous continuity***. Most of the inlet portion is finely trabeculated with ***anterolateral*** (anterior) and ***posteromedial*** (posterior) ***papillary muscles*** connecting to the mitral valve leaflets via ***tendinae chordae***.

Atrial septum

Although at first glance it looks as though the lateral RA wall between the IVC and SVC is the atrial septum, the true septum is virtually confined to the ***fossa ovalis***. The superior rim of the fossa is commonly referred to as the ***septum secundum***, but it is really an infolding between the right atrium and the pulmonary veins.

Ventricular septum

The ventricular septal surfaces are asymmetrical firstly because of the infundibular portion of the RV, secondly because of the differing long axis of RV (vertical) and LV (oblique), and the pressure differential between RV and LV. It is made up of a ***muscular septum, a membranous septum***, and the ***atrioventricular septum***. The atrioventricular septum lies between the right atrium and the left ventricle. The ***AV node*** lies in the atrial septum adjacent to the junction between the membranous and muscular portions of the atrioventricular septum, and the ***bundle of His*** passes toward the ***right fibrous trigone*** between these components.

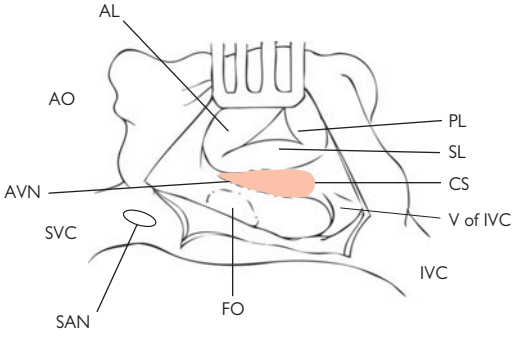


Fig. 2.4 Interior of the right atrium: AO aorta, AL anterior leaflet, PL posterior leaflet and SL septal leaflet of tricuspid valve, AVN atrioventricular node, SVC superior vena cava, FO foramen ovale, IVC inferior vena cava, V of IVC Esutacian valve, CS coronary sinus, T tendon of Todaro, TK triangle of Koch.

Cardiac valves

Key facts

The valves occupy a central position between the mitral and tricuspid valves, with the pulmonary valve slightly superior, anterior, and to the left. The valves are thin folds of endocardium covered by endothelium. All cardiac valves share the same basic histological structure:

- Thin ventricular layer (radially aligned collagen fibers).
- Central spongiosa (loosely arranged collagen and proteoglycans).
- Thick fibrous layer (densely packed fibers arranged parallel to the leaflet edge) which provides structural integrity and stability.
- Thin upper collagenous layer.
- The aortic and pulmonary valve leaflets are nearly avascular: they are thin enough to be perfused from surrounding blood, whereas the mitral and tricuspid valves contain a few capillaries.

Tricuspid valve

This has an anterior, a septal, and a posterior leaflet. The orifice is triangular, larger than the mitral valve, and the annulus is much less well defined. The leaflets and chordae are thinner than the mitral valve. The chordae to the largest anterior leaflet arise from the anterior and medial papillary muscles. The posterior leaflet is the smallest, most inferior, and is usually scalloped. The chordae to the septal leaflet arise from the posterior and septal papillary muscles. The conduction system is closely related to the septal leaflet (Fig. 2.5).

Pulmonary valve

This is made of right, left, and anterior (non-septal) leaflets. The structure of the pulmonary valve is similar to the aortic valve, with three differences. Firstly, the valve leaflets are flimsier than the aortic valve; secondly, coronary arteries do not originate from the sinuses; and thirdly, the valve is not in continuity with the anterior tricuspid valve leaflet.

Mitral valve

This bicuspid valve has a large anterior (aortic or septal) leaflet and a small posterior (mural or ventricular) leaflet. The leaflet area is much greater than the valve area, allowing a large area for coaptation, distinguished as a rough zone. The larger anterior leaflet inserts on only a third of the annulus, through which it is fibrous continuity with the left and half of the non-coronary cusp of the aortic valve. The smaller posterior leaflet inserts into 2/3 of the annulus and is scalloped. Each leaflet is segmented into three for the purposes of nomenclature (Fig. 2.5). The chordae tendinae insert into both leaflets from the anterolateral and posteromedial papillary muscles. There are three orders of chordae: 1st (marginal) insert onto the leaflet free margin, 2nd order insert a few millimeters further back, and 3rd (basal) insert at the base of the posterior leaflet only.

Aortic valve

This tricuspid valve is in fibrous continuity with the anterior leaflet of the mitral valve and the membranous septum. The free edge of each cusp is thickest and at the midpoint of each free edge is a fibrous nodulus arantii,

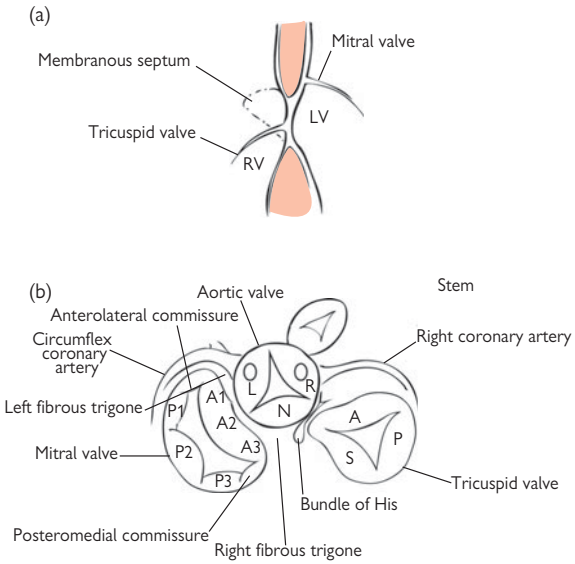


Fig. 2.5 (a) Atrioventricular junction: note the septal atttachment of the tricuspid valve is lower than the septal attachment of the mitral valve. (b) Cardiac valves and their relationships viewed from above, with the atria removed. Note how the commissure of the left (L) and the noncoronary (N) sinuses of the aortic valve points at the midpoint of the anterior leaflet of the mitral valve (A2). The circumflex coronary artery is particularly close to the posterior mitral annulus in the P3, P2 segment. The right coronary artery is close to anterior tricuspid annulus. The bundle of His lies in the membranous septum near the right fibrous trigone located at the junction of the right (R) and the non-coronary cusp (N), extending to the antero-septal commissure of the tricuspid valve.

bordered on either side by crescent-shaped lunula which form the region of coaptation. The aortic sinuses (sinuses of Valsalva) are dilated, relatively thin-walled, pockets of the aortic root, two of which give rise to the coronary arteries. Because of the shape of the cusps the annulus is crown shaped. The cusps are called right, left, and non-coronary, based on the origin of the coronary arteries (Fig. 2.5).

Fibrous trigones

The mitral and tricuspid annuli are in fibrous continuity with each other and the membranous septum, forming the fibrous skeleton of the heart. The central fibrous trigones lie between the mitral and tricuspid valves (Fig. 2.5): the right trigone sits between the mitral and tricuspid annuli, the non-coronary cusp of the aortic valve and the membranous septum. The left fibrous trigone lies between the ventriculoaortic junction and the mitral annulus. The bundle of His pierces the right fibrous trigone.

Coronary arteries and veins

Key facts

The coronary arteries are two main distributions (right and left), supplied by three vessels; ***right coronary artery*** (RCA), ***left anterior descending*** (LAD), ***and circumflex artery*** (Cx) (Fig. 2.6).

Left main stem

The LMS arises from the ostium of the left sinus of Valsalva, travels between the pulmonary trunk anteriorly and the left atrial appendage to the left AV groove, dividing after 1–2cm into LAD, Cx, and occasionally a third artery: the intermediate (Int).

Left anterior descending

The LAD runs down the anterior interventricular groove to the apex of the heart, usually extending round the apex to the posterior interventricular groove and the territory of the PDA. A variable number of ***diagonals*** are given off over the anterior surface of the LV, small branches supply the anterior surface of the RV, and superior ***septals*** are given off perpendicularly to supply the anterior 2/3 of the interventricular septum. The first septal is the largest. Some of the RV branches anastomose with infundibular branches of the proximal RCA: ***loop of Vieussens***.

Circumflex

The Cx originates at 90° from the LMS and runs medially to the LA appendage for 2–3cm, continuing in the posterior left AV groove to the crux of the heart. In left dominant hearts (5–10%) the Cx turns 90° into the posterior interventricular groove to form the posterior descending artery (PDA). In 85–90% of hearts the PDA arises from the RCA (right dominant). About 5% of hearts are co-dominant. A variable number of ***obtuse marginals*** (OM) arise from the Cx to supply the posterior LV. They are frequently intramuscular. The first branch of the Cx is the ***AV nodal artery*** in 45% which courses round the LA near the AV groove.

Right coronary artery

The RCA arises from an ostium in the right sinus of Valsalva, gives off an ***infundibular branch*** and then a branch to the ***SA node*** early, and runs immediately into the deep right AV groove where it gives off ***RV branches*** to the anterior RV wall. The ***acute marginal*** is a large branch which crosses the acute margin of the heart to travel to the apex. In right dominant hearts the RCA reaches the crux of the hearts where it turns 90° to form the ***PDA***, which runs towards the apex in the posterior interventricular groove. ***Inferior septals*** which supply the inferior 1/3 of the interventricular septum arise at 90° from the PDA. The ***AV node artery*** is given off by the RCA in 55% at the crux. One or more ***right posterolateral branches*** (RPLSs) supply the LV.

Venous drainage of the heart

The coronary sinus runs in the posterior AV groove draining into the RA with the great cardiac vein. Thebesian veins empty directly into the heart.

Variations in coronary anatomy

- Right dominant 90%, left dominant 5%, co-dominant 5%.
- Absent LMS with separate ostia for the Cx and the LAD (1%).
- LAD or RCA represented by 2 separate parallel vessels (2%).
- SA node artery arising from the Cx (2–3%).
- SA node artery may travel round the SVC clockwise or anticlockwise, or bifurcate and travel in both directions (Fig. 2.6b).
- AV node artery from RCA (55%) from Cx (45%).

Variations causing ischemia

- *ALCAPA* (anomalous left coronary artery from the pulmonary artery <0.01%) and coronary ostial atresia cause severe ischemia. Untreated, most patients die in infancy. If large collateral network patients may survive to adulthood: treatment for adults is CABG.
- *ACAOS* (anomalous origin of a coronary artery from the opposite sinus), e.g., origin of the LMS from the RCA then passing between RVOT and aorta (or anterior to the PA, or posterior to the aorta) may cause intermittent ischemia and sudden death during or shortly after exercise, due to increased blood flow and pressure in these arteries during exercise, slit-like ostium. CABG traditional approach.

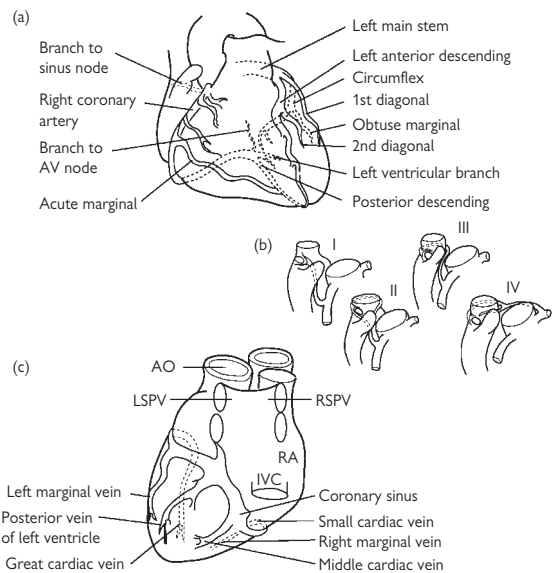


Fig. 2.6 (a) Coronary arteries. (b) Variations in the sinus node artery, which may arise from the right coronary artery (60%) and encircle the base of the superior vena cava in a clockwise (I), anticlockwise (II) or both (III) directions, or arise from the left coronary artery (40% of cases) (IV). (c) Coronary veins.

Lung anatomy

Lobes

- The right lung has three lobes: an upper, a middle, and a lower lobe.
- The smaller left lung has two lobes: an upper, and a lower lobe.

The lingular segment of the left upper lobe corresponds to the right middle lobe. An azygos lobe is found in up to 1% of lungs in the right upper lobe. It is formed from varying portions of the apical and/or posterior segments by an aberrant azygos vein and its pleural mesentery. The fissure is visible on the chest radiograph as an inverted comma.

Fissures

- The right lung has two fissures: an oblique and a transverse fissure.
- The left lung has one fissure: the oblique fissure.

The oblique fissure separates the upper lobe (and middle lobe on the right) from the lower lobe on each side. The transverse fissure separates the upper lobe from the middle lobe on the right. It is often incomplete.

Bronchopulmonary segments

- A bronchopulmonary segment is a portion of lung which functions as an individual unit, having its own artery and bronchus. There are 18.
- The right lung is divided into 10 bronchopulmonary segments (Fig. 13.2).
- The left lung is divided into 8 bronchopulmonary segments (Fig. 13.2).

Tracheobronchial anatomy

The trachea commences at the lower border of the cricoid cartilage (C6) and terminates at the carina (T7) dividing into right and left main bronchi.

- Right main bronchus is 1.2cm long.
- After the upper lobe bronchial orifice arises, the bronchus intermedius (BI) is given off. It is ~2cm long.
- The middle lobe bronchus arises from anterior surface of the BI, and the superior segmental bronchus to the lower lobe arises from the posterior wall of the distal BI below the middle lobe.
- The BI then divides to form the basal segmental bronchi.
- The left main bronchus is 4–6cm long.
- After the upper lobe bronchial orifice the left main bronchus continues as the lower lobe bronchus: its first branch is the superior segmental bronchus, arising posteriorly.
- The lower lobe bronchus divides to form the basal segmental bronchi.

Tracheal anatomy

- The average adult male trachea is 10–13cm in length, measured from the lower edge of the cricoid ring to the carina.
- 1/3 of the adult trachea is above the sternal notch, with the balance an intrathoracic structure.
- In the absence of congenital tracheal stenosis, the only complete tracheal ring is the cricoid cartilage; all other rings are incomplete ring of cartilage with a posterior membranous wall.

- There are 18–22 D-shaped partial rings. The cartilaginous rings are softer in the pediatric population, and can become calcified with age.
- The arterial supply of the trachea is segmental, or entering via lateral pedicles. Dissection of the trachea should be limited to anterior and posterior planes. In the absence of tracheal resection, circumferential dissection should be limited to 1–2cm to avoid ischemic necrosis.

Upper half of trachea

- Tracheoesophageal branches of the inferior thyroid artery.
- Rarely, branches from the subclavian arteries.

Middle to lower trachea and carina

- Branches from the supreme intercostal artery, subclavian artery, right internal mammary artery, innominate artery.
- Branches from the bronchial arteries.

Pulmonary vessels

Pulmonary trunk

- This arises intrapericardially from the infundibulum of the RV. It continues superioposteriorly to the left for approximately 5cm as the main pulmonary artery. It bifurcates just below the aortic arch.

Right pulmonary artery

- Passes posterior to aorta and SVC. Initially anterior and inferior to right main bronchus, and superior and posterior to the superior pulmonary vein. It gives off three branches: (1) truncus anterior branch to upper lobe, (2) middle lobe artery arises anteromedially and distal to the truncus anterior, (3) posterolaterally at a similar level the superior segmental branch to the lower lobe arises. The artery distal to this becomes the common basal trunk which divides to form the basal segmental vessels.

Left pulmonary artery

- This passes laterally and posteriorly under the aortic arch before giving off the upper lobe branches, usually four. The most distal branch to the upper lobe is the lingular branch. Posteriorly at the level of the fissure the branch supplying the superior segment of the lower lobe arises. Distal to this the vessel becomes the common basal trunk and subsequently divides into two main vessels supplying the lower lobe segments.

Pulmonary veins

- There are two pulmonary veins on each side, a superior and an inferior. The veins drain into the LA. At the hilum the right superior pulmonary vein is anteroinferior to the artery. It usually drains the upper and middle lobes. Inferoposteriorly lies the inferior vein. It drains the lower lobe and rarely the middle lobe. The middle lobe may have its own vein which enters the right atrium separate to the superior pulmonary vein. At the hilum the left superior vein is anteroinferior to the artery. It drains the upper lobe including the lingular segments. Occasionally the lingular segments drain into the inferior vein.

Bronchial arterial system

- This arises from systemic circulation, from various arterial origins. Drainage is into the pulmonary venous system, azygos, and hemiazygos veins.

Anatomy of the esophagus

The esophagus is a continuation of the pharynx, arising at the level of C6 and the cricoid cartilage, and ending at the stomach at T11. It attaches superiorly to cricoid cartilage and inferiorly to the diaphragm. The average length from cricopharyngeus muscle to gastroesophageal junction is 25cm in males and 23cm in females. The average length from incisor teeth to cricopharyngeus is 15cm in males and 14cm in females (Fig. 13.4). The length from incisor teeth to middle indentation is 24–26cm. These measurements are important landmarks when accurately describing the position of lesions in the esophagus at esophagoscopy. There are three regions of narrowing/indentations along its length:

- At the origin, caused by cricopharyngeus.
- The middle indentation, caused by aortic arch and left main bronchus.
- At the distal end, caused by the lower esophageal sphincter at the diaphragm.

Relationships in the thorax

The esophagus enters the thorax through the thoracic inlet. It is intimately related to the prevertebral fascia down to the tracheal bifurcation. It moves slightly to the right just above the bifurcation, then passes posterior to the pericardium in line with the left atrium to reach the esophageal opening in the diaphragm. At the diaphragm the thoracic duct lies behind the esophagus. As it ascends in the thorax it gradually moves to the left of the esophagus. It passes dorsal to the aortic arch and drains into the venous system at the union of the left subclavian and internal jugular veins (see Fig. 2.7).

Musculature of the esophagus

- The upper part consists of striated muscle fibres only. Smooth muscle fibres increase in number on descending the esophagus until they constitute the entire musculature. There is an outer longitudinal and an inner thicker circular layer of muscle, allowing effective peristalsis.

Arterial supply

- Cervical esophagus: inferior thyroid artery.
- Thoracic esophagus: bronchial arteries. Two esophageal branches also typically arise directly from the descending aorta.
- Abdominal esophagus: inferior phrenic and esophageal branches of inferior gastric arteries.
- There is an extensive intramural arterial network allowing mobilization without devascularization.

Venous drainage

- A submucosal venous plexus drains into periesophageal venous plexus which then forms the esophageal veins. In the cervical region the esophagus drains into the inferior thyroid vein, in the thorax into the azygos, hemiazygos, and bronchial veins, in the abdominal region into the coronary vein. There is continuity between esophageal and stomach venous networks, important in portal hypertension.

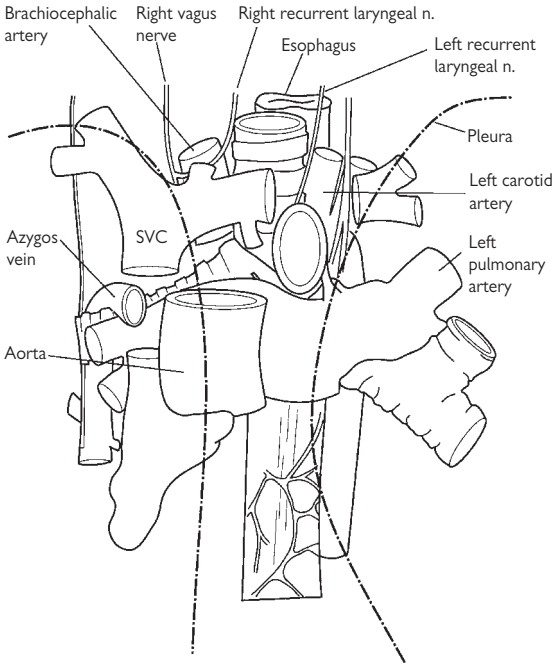


Fig. 2.7 Relationships of trachea, esophagus, and great vessels in the mediastinum.

Nerve supply

Parasympathetic supply

Vagus nerve–recurrent laryngeal nerves superiorly, esophageal plexus inferiorly.

Sympathetic supply

The pharyngeal plexus receives fibres from the superior cervical ganglion. Postganglionic fibres from the cervical and thoracic sympathetic chain then reach the esophagus. The distal esophagus also receives sympathetic fibres from the celiac ganglion.

Thoracic vessels

Ascending aorta

The ascending aorta lies in the middle mediastinum, beginning at the base of the LV. This ***ventriculoaortic junction*** is also known as the ***aortic annulus*** (Fig. 2.8). The corona-shaped annulus is in fibrous continuity with the anterior leaflet of the mitral valve and the membranous septum (Fig. 8.9). Distal to the aortic annulus are three thinner-walled bulges: the ***aortic sinuses***. The ring-shaped junction between the aortic sinuses and the remaining aorta is known as the ***sinotubular junction*** *(STJ)* (Fig. 2.8). The ***aortic root*** is the ascending aorta from annulus to STJ. The ascending aorta carries on for ~5cm, curving forwards and right behind the left half of the sternum to the 2nd left costal cartilage.

Relations

- Pericardium, thymic remnants, areolar tissue, innominate vein, lungs and pleura, sternum anteriorly.
- LA, right pulmonary artery, right main bronchus posteriorly.
- SVC and RA to the right, LA and PA to the left.

Branches

- Right and left coronary arteries.

Arch of aorta

The whole of the aortic arch lies in the superior mediastinum. The aortic arch begins behind the right half of the manubrium, at the 2nd right costal cartilage. It curves posteriorly and to the left, ending at the level of the 2nd left costal cartilage, or 4th thoracic vertebra.

Relations

- Four nerves anteriorly: left phrenic, left vagus, left cardiac branch of vagus nerve, and left cervical branch of sympathetic chain.
- Four nerves posteriorly: left recurrent laryngeal nerve (given off by left vagus nerve, looping back under the aortic arch), deep cardiac plexus, right vagus nerve and right phrenic nerve (separated from the aortic arch by the trachea and SVC respectively).
- Four structures posteriorly: trachea, esophagus, vertebral bodies, thoracic duct.

Branches

Brachiocephalic artery, left common carotid artery, left subclavian artery, occasionally thyroidea IMA artery, bronchial arteries.

Descending thoracic aorta

The descending aorta lies in the posterior mediastinum. It begins at the level of the 4th thoracic vertebra, ending at the 12th thoracic vertebral border, where it passes through the diaphragm to become the abdominal aorta. It goes from being to the left of the vertebral column superiorly, to being directly anterior to it at the level of the diaphragm.

Relations

- Left pulmonary hilum, LA, esophagus, diaphragm anteriorly.
- Vertebral column and hemiazygos veins posteriorly.
- Azygos vein, thoracic duct, and lower esophagus to the right.
- Pleura and left lung to the left.

Branches

Pericardial branches, right and left bronchial arteries, which run posteriorly on the bronchi, four or five esophageal arteries, mediastinal and phrenic branches, nine paired posterior intercostal arteries (see also 📖 p36), paired subcostal arteries.

Inferior vena cava

Conveys blood to the RA from all structures below diaphragm. Formed by the junction of the common iliac veins, ascends anterior to the vertebral column, to the right of the aorta. It traverses the posterior surface of the liver via a deep groove, piercing the tendinous part of the diaphragm and pericardium to enter the RA. Collateral circulations exist between epigastric, internal thoracic, hemiazygos, azygos, and pelvic veins.

Superior vena cava

Returns blood from upper half of the body. Formed by junction of the brachiocephalic veins, and receives azygos vein before entering the pericardium. It is valveless. Descends behind the 1st to the 3rd right costal cartilage. The right pulmonary hilum is posterior, the right phrenic nerve is right lateral, and the trachea and right vagus nerve lie posteromedial.

Azygos vein

Returns blood from the posterior thoracic wall, esophagus, bronchi, pericardium and mediastinal lymph nodes. Variable origin. Ascends anterior to thoracic vertebrae in the posterior mediastinum to T4, where it arches over the right pulmonary hilum ending in the SVC, which it enters 1–2cm distal to junction with RA posteriorly.

Hemiazygos and accessory hemiazygos veins

The hemiazygos ascends on the left like the azygos vein, crossing the vertebral column posterior to the aorta at a variable level to end in the azygos vein. The accessory hemiazygos vein descends to the left of the upper vertebral column, crossing at T7 to join the azygos vein.

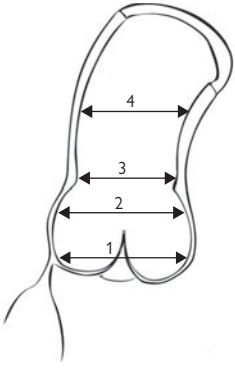


Fig. 2.8 Aortic root dimensions. 1) Annulus, 2) sinuses, 3) sinotubular junction, 4) ascending aorta. Drawn by Marcia Williams.

Vascular anatomy and pathology

Peripheral cannulation sites

Axillary artery and vein

The axillary artery, which is less prone to atherosclerosis or aneurysm than the aorta or femoral artery, is located deep to the deltopectoral groove (Fig. 2.9). Vein lies caudal to artery. The ***subclavian artery*** becomes the axillary artery at the 1st rib lateral margin. Divided into three parts by ***pectoralis minor***. 1st part proximal to pectoralis minor, has one branch: superior thoracic artery. 2nd part posterior to pectoralis minor has two branches: thoracoacromial and lateral thoracic artery. 3rd distal part has three branches: subscapular (largest), anterior and posterior circumflex humeral arteries. Three cords of ***brachial plexus*** wrap around 2nd part: lateral cord contains C5–C7, medial cord (ulnar nerve) contains C8–T1, posterior cord (axillary and radial nerve) contains C5–T1 contributions.

Femoral artery and vein

Located at midinguinal point: femoral nerve laterally, femoral vein medial. The ***profunda femoris*** is its largest branch originating from its lateral side in the femoral triangle, and running posteriorly. More vasospastic in young patients, and more atherosclerotic in elderly than aorta or axillary artery.

Conduit

Internal mammary artery

Arises inferiorly from left ***subclavian*** artery 2cm above the sternal clavicle, opposite the thyrocervical trunk. Descends anteromedially behind the clavicle and ***brachiocephalic veins*** and 1st costal cartilage, between two venae comitantes. Crossed by ***phrenic nerve*** from lateral to medial as it enters thorax. Descends vertically over the first six costal cartilages, bifurcating into the ***superior epigastric*** and ***musculophrenic arteries*** at the xiphisternal junction. Separated from pleurae after 3rd rib by fascia and transversus thoracis. ***Intercostal nerves*** cross it anteriorly in each rib space. Pericardiacophrenic artery is proximal branch. Two anterior intercostal branches per rib space. RIMA is more medial proximally.

Radial artery

Smaller than ulnar artery, but a more direct continuation of ***brachial artery***, beginning medial to the neck of the radius, 1cm distal to the elbow crease, 1–2cm distal to the bicipital aponeurosis. Overlapped by ***brachioradialis*** for the proximal 2/3 (Fig. 7.6). Distal 1/3 covered only by skin and fascia. Overlies (from proximal to distal) the tendon of biceps, supinator, distal attachment of pronator teres, radial head of flexor digitorum superficialis, flexor pollicis longus, pronator quadratus, and distal radius. Superficial branch of ***radial nerve*** is closely related to the radial artery in its middle 1/3, and filaments of ***lateral cutaneous nerve*** of forearm run over it as it curves around carpus. The 1st branch is ***recurrent radial artery*** which supplies elbow joint. Muscular branches are distributed throughout. The ***deep palmar arch*** via which it anastomoses with the ulnar artery is absent in 3%. ***Median nerve*** lies deep to flexor digitorum superficialis. The lateral cutaneous nerve of the forearm lies lateral and parallel to the standard incision (Fig. 7.6).

Long and short saphenous veins

Continuation of medial marginal vein, 2cm anterior to ***medial malleolus***, crossing distal 1/3 of medial surface of tibia ascending just posterior to the tibial border to knee (Fig. 7.1a). Here it lies 2cm posteromedial to medial tibial and femoral condyles. Ascends medial aspect of thigh traversing the saphenous opening to empty into femoral vein. The saphenous opening is located ~3cm inferolateral to pubic tubercle. Receives multiple tributaries, particularly around knee, from superficial and deep veins. The vein may be duplicated in the lower leg, and contains 10–20 valves. The ***saphenous nerve*** distally and the ***median femoral cutaneous nerve*** proximally are closely associated, as is the saphenous branch of the genicular artery at the knee. ***Short saphenous vein*** commences 1cm posterior to lateral malleolus running lateral to Achilles tendon, ascending in midline with ***sural nerve***, to join popliteal vein in popliteal fossa (Fig. 7.5a).

Conduit pathophysiology and patency

- Conduit patency (📖 p302) is dictated by three main factors: (1) technical factors (📖 p323), (2) targets, i.e., distal run-off (📖 p323) and severity of stenosis being bypassed (📖 p323), and (3) intrinsic qualities of the conduit
- There are several intrinsic differences between the IMA, radial artery, and SVG that may contribute to the different patency rates observed:
  - IMA has non-fenestrated elastic lamina that may reduce cellular migration and intimal hyperplasia (📖 p323) whereas the radial artery and SVG have fenestrated elastic lamina.
  - IMA thinner and less vasoreactive muscular layer than radial artery (it does not vasoconstrict in response to norepinephrine), and also reduced proliferative response: so less prone to early vasospasm.
  - Unlike SVG and radial, IMA adapts in response to flow over time.
  - SVG has a lower propensity to spasm than arterial conduit, but reduced endothelial resistance to thrombosis and atherosclerosis.

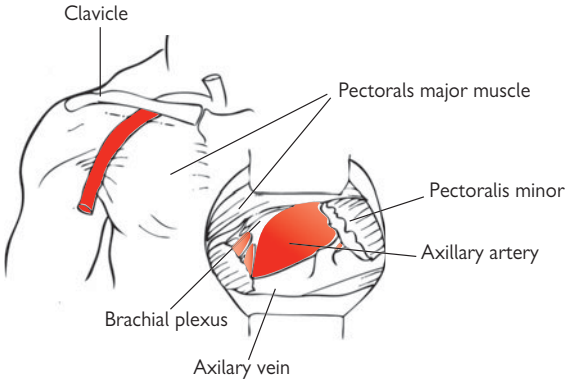


Fig. 2.9 The axillary artery.

Myocardium

Specialized cardiac tissues

The heart is composed of three cardiac muscle types: atrial muscle, ventricular muscle, and specialized excitatory and conductive muscle.

Atrial and ventricular muscle

The basic unit of the cardiac contractile apparatus is the ***sarcomere***. This is the part of a ***myofibril*** lying between adjacent Z discs. Cardiac muscle fibres are composed of hundreds of myofibrils containing several thousand ***actin*** and ***myosin filaments***, the protein polymers responsible for muscle contraction. Actin filaments are made of three proteins: actin, ***troponin***, and tropomyosin. The ends of the actin filaments connect to a strip of protein filaments called the ***Z disk***, which passes from one myofibril to another, connecting to all the myofibrils across the muscle fiber. This arrangement gives cardiac muscle and skeletal muscle its striated appearance. Myofibrils are suspended inside the muscle fiber in a matrix called sarcoplasm, which contains a high concentration of K^+, Mg^{2+}, PO_4, protein enzymes, mitochondria and an extensive sarcolemma. Cardiac muscle differs from skeletal muscle in several ways:

- ***Rhythmicity*** (see 📖 p58) and resistance to tetany and fatigue.
- Cardiac myofibrils interconnect to form a network or ***synctium***.
- Individual muscle cell membranes, called ***intercalated disks***, are extremely permeable, allowing the action potential to move unhindered between muscle cells, and across the lattice.
- The action potential does not spread instantly from atria to ventricles because the heart is composed of two separate syncytiums, an atrial and a ventricular one, separated from each other by fibrous tissue which surrounds the valve orifices: the delay between atrial and ventricular contraction is critical for cardiac function.
- Normally the only way that an action potential can be conducted from atria to ventricles is via the specialized conducting pathways.

Excitatory and conducting tissue

These muscle fibres contract very weakly, because they contain few contractile fibrils. They generate and conduct the cardiac action potential. Fig. 2.4 shows the ***sinus node (sinoatrial or SA node)*** in which the normal rhythmic impulse is generated, the internodal pathways connecting it to the ***AV node*** which slows conduction, and the ***bundle of His*** which conducts the impulse to the ***Purkinje fibres*** and the ventricles. The anatomy of the conducting system is described on 📖 p43.

Excitation

Many cardiac fibres generate automatic rhythmical impulses. Normally the sinus node generates the fastest impulse rate, therefore controlling the heart rate. Sinus node self-excitation depends on: (1) selective membrane permeability to Na ions, and (2) three types of voltage-gated ion channels (fast Na channels, slow Ca/Na channels, and K channels).

- The negative resting potential of myocytes membranes (−80mV) is mostly due to the difference in K^+ ion concentration inside the cell

(140mEq/L) compared to outside (4mEq/L) partially offset by the opposing difference in Na^+ concentrations inside the cell (14mEq/L) compared to outside (142mEq/L).
- The sodium–potassium pump contributes about −4mEq/L to the resting potential by pumping out three Na^+ for every two K^+ pumped in.
- The conductive cells of the heart are normally relatively impermeable to Na^+, which reduces the negative resting potential slightly.
- The sinus node cells are fairly permeable to Na^+, so the negative resting potential is offset to a much greater amount: the resting potential of the sinus node cells is less negative at –55 to –60mV than it is in ventricular fibers at –85 to –90 mV.
- The ***action potential*** (an electrical nerve or muscle impulse) is generated when the slow leakage of Na^+ into the cell lowers the ***resting potential*** to –40mV.
- The calcium–sodium channels are activated, causing rapid influx of Ca^{2+} and Na^+: these channels close 100ms later, stopping the influx of positive ions.
- The membrane is now ***depolarized***: the potential is no longer negative.
- At this point the voltage-gated K^+ channels open, and large amounts of K^+ leave the cell, terminating the action potential.
- This continues for a longer time than depolarization: the membrane is temporarily much more negative (hyperpolarization): the K^+ channels eventually close, and Na^+ diffuse in slowly, returning the membrane to resting and then threshold potential.

Conduction

An action potential elicited at one point of a conducting tissue, normally excites adjacent portions, resulting in ***propagation*** of the action potential. This occurs at the same speed in all directions. The specialized conducting tissues of the heart are designed to propagate the impulse in such a way that both atria contract a short time before the ventricles. The impulse is delayed as it passes through the AV node and the bundle of His, firstly because the conducting fibers are much smaller there, secondly because the threshold membrane potentials are much less negative, and thirdly because there a fewer gap junctions connecting fibers together. AV conduction is normally one way, preventing re-entry of cardiac impulses via this route. Purkinje fibers conduct so fast that the cardiac impulse is transmitted throughout each ventricle almost simultaneously. Abnormal pacemakers are described on 📖 p484.

Contraction

The action potential travels along muscle fibers in the same way. The action potential depolarizes the muscle fiber membrane and causes the sarcolemma to release large quantities of calcium into the myofibrils. The calcium ions initiate attractive forces between the actin and myosin filaments, causing them to slide together, resulting in muscle contraction. This process requires energy derived from the conversion of adenosine triphosphate (ATP) to adenosine diphosphate (ADP) in the myosin molecule head. Like any muscle, cardiac muscle generates its maximum force of contraction from its resting length.

The electrocardiogram

Key facts

The action potential, already described on 📖 p56, is shown in Fig. 2.10a for the nodal tissues and non-nodal tissues of the heart. As the cardiac impulses pass through the heart electrical currents spread into surrounding tissues and can be detected by electrodes on the skin surface. The voltages recorded are small: the QRS complex is about 3–4mV compared to the monophasic action potential of 100mV recorded by an electrode inserted into cardiac muscle. Note that no potential at all is recorded in the electrocardiogram (EKG) when the ventricular muscle is completely depolarized or completely polarized. Current only flows during depolarization and repolarization.

The normal EKG

The normal EKG (Fig. 2.10b) is composed of a P wave, a QRS complex, and a T wave. The ***P wave*** is caused by depolarization of the atria prior to contraction, the ***QRS complex*** is caused by depolarization of the ventricles prior to contraction (and obscures the atrial T wave caused by atrial repolarization), and the ***T wave*** is caused by ventricular repolarization. The heart rate is estimated from the ***RR interval***. Normal heart rhythm (sinus rhythm) is implied by a normal P wave followed by a QRS complex, with a constant PR interval and a constant RR interval.

- In standard recordings 1 small square of EKG paper = 0.04s.
- The P wave normally lasts for 0.12s (3 small squares).
- The PR interval is normally 0.16s (4 small squares).
- The QRS complex is usually <0.12s (3 small squares).
- The PR segment occupies the ***isoelectric point***, and under normal conditions so does the ST segment.
- Inverted waves in certain leads are abnormal. Interpretation of EKGs is outlined on 📖 p212.

Understanding EKG leads

The standard bipolar limb leads (I, II, and III) are not single wires connecting from the body, but a combination of two wires and their electrodes which make a complete circuit through the tissue between them to the electrocardiograph (Fig. 2.10c).

- The augmented (a) limb leads (aVR, aVL, and aVF) are unipolar limb leads where two of the leads are connected to the negative terminal of the EKG and one is connected to the positive.
- The positive terminal may be the right arm (aVR), the left arm (aVL), or the left leg (aVF).
- The six chest leads (V1–V6) consist of one wire connected to the body in the positions shown, and an indifferent wire connected to all three limb leads.
- The chest leads record the potential of the cardiac muscle directly beneath them, and minute changes in particularly the anterior ventricular wall appear in these leads.

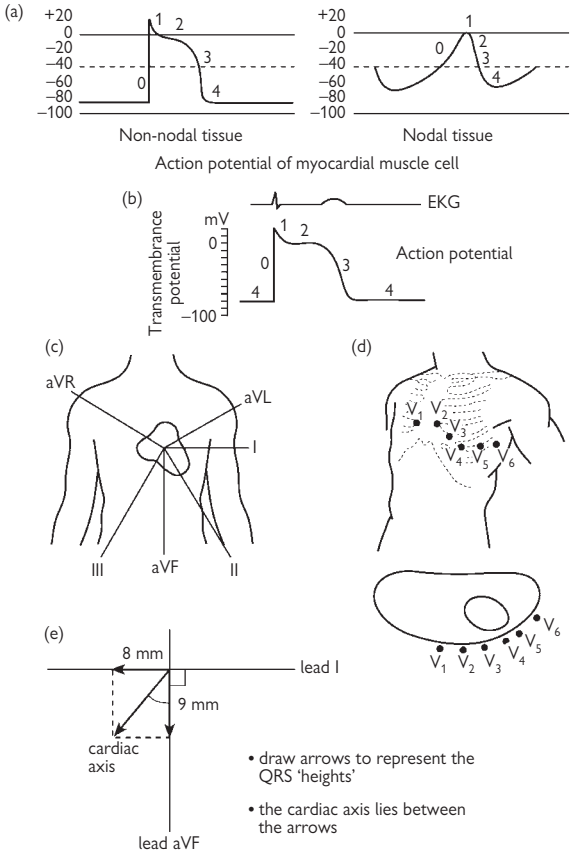


Fig. 2.10 (a) Monophasic action potentials of nodal tissue, and non-nodal tissue, and their relationship to the EKG. (b) ECK superimposed on action potential (c) Axes of the three bipolar and three unipolar limb leads. The axis of a lead is the direction from positive to negative electrode. Lead I is recorded from two electrodes that lie in the horizontal position with respect to the chest: the axis is 0. Lead aVF is 90°. The size of current recorded and the direction (positive or negative) in a lead can be recorded. (d) The chest leads. (e) The mean electrical axis of the QRS complex is normally 60°. The axis of the QRS complex can be calculated by adding the QRS complex in any two leads.

Cardiac function

The cardiac cycle

The cardiac cycle is the period from the beginning of one heart beat, normally initiated by spontaneous generation of an action potential in the sinus node, to the beginning of the next (Fig. 2.11). Blood normally flows continually from the great veins into the atria: about 75% flows directly into the ventricles before the atria contact. Atrial contraction normally causes an additional 25% filling of the ventricles (***atrial kick***).

Ventricular function

Filling phase: during ***ventricular systole*** closure of the AV valves, combined with continuous inflow of blood from the great veins, means that large amounts of blood collect in both atria. As soon as systole is finished and ventricular pressures fall below atrial pressure (5), the AV valves open (v) allowing blood to flow rapidly into the ventricles. This is the period of ***rapid ventricular filling*** (6), and it lasts the first 1/3 of diastole. During the middle third additional filling is directly from the great cardiac veins (7). During the final 1/3 of diastole the atria contract (1) accounting for a further 25% of ventricular filling. Filling of each ventricle is 110–120mL. Diastole is an active process, requiring energy.

Isovolumetric contraction: as the ventricles start to contract against the closed semilunar valves the AV valves are forced closed (c). Intraventricular pressure builds until it is greater than the pressure in the great arteries (2), causing the semilunar valves to open. There is apex-to-base shortening and circumferential elongation. LV volume is unchanged.

Ejection phase: once the semilunar valves open blood empties into the great arteries, 70% during the first 1/3 of the ejection period (3) (***rapid ejection phase***) and the remaining 30% during the last 2/3 (4) (***slow ejection phase***) or ***isotonic contraction*** phase.

Isovolumetric relaxation: ventricular diastole begins suddenly at the end of systole, and intraventricular pressures fall rapidly (5). The semilunar valves close as intraventricular pressure falls below the pressure in the great arteries. The ventricular muscle continues to relax without change in volume, until the AV valves open (v) as ventricular pressures fall below atrial pressure, and the cycle begins again.

Definitions

- ***End-diastolic volume*** (EDV) = largest ventricular volume (~120mL) immediately prior to the beginning of systole.
- ***End-systolic volume*** (ESV) = smallest ventricular volume (~40mL) immediately before the beginning of diastole.
- The ***stroke volume*** = amount ejected per cycle (120 – 40 = 80mL).
- The ***ejection fraction*** = percentage ejected (80/120mL = 66%).
- The ***cardiac output*** = stroke volume times the heart rate. The normal heart can increase cardiac output by up to 6x depending on demand, by decreasing ESV, increasing EDV, and increasing heart rate.
- ***Compliance*** is unit increase in volume/unit increase in pressure. It is a measure of active and passive relaxation and passive stretch.

Pressure–volume loops

Pressure–volume loops (Fig. 8.3) are derived by filling the heart with increasing amounts of blood and then measuring the pressure immediately before systole to obtain the diastolic curve; and by preventing any outflow of blood and measuring the maximum intraventricular pressure to obtain the systolic curve. Normally the heart operates within a much smaller area. The area enclosed within the pressure–volume loop is equal to the net ***external work output*** of the ventricle. The slope of the diastolic curve is inversely proportional to ***compliance*** (the ability of the ventricle to relax). The pressure–volume loops associated with valvular heart disease are shown on 📖 p375.

Preload and afterload

- ***Preload*** is the volume of blood in the ventricle at the end of diastole (the EDV). It is affected by circulating volume, LV compliance, the length of diastole and atrial contraction.

Afterload is the resistance to ventricular ejection. It is affected by the outflow tract area, peripheral resistance and wall stress. Manipulating preload and afterload is described on 📖 pp186–187.

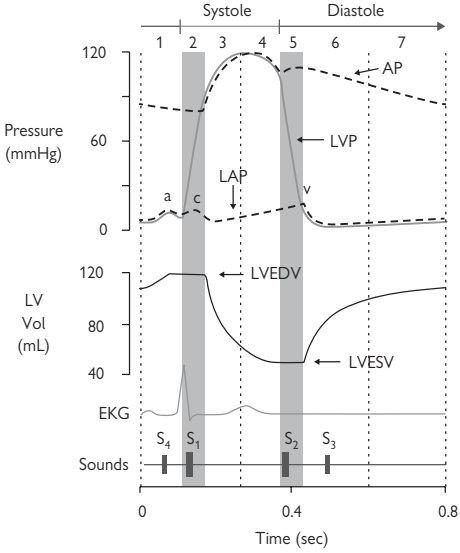


Fig. 2.11 The cardiac cycle. AP, aortic pressure; LVP, left ventricular pressure; LAP, left atrial pressure; LVEDV, left ventricular end–diastolic volume; LVESV, left ventricular end systolic volume. 1, atrial contraction; 2, isovolumetric contraction; 3, rapid ejection; 4, reduced ejection; 5, isovolumetric relaxation; 6, rapid filling; 7, reduced filling.

Circulation

Key facts

Blood flow to body tissues and cardiac output is controlled in relation to oxygen demand. BP control under normal circumstances is regulated so that it is nearly independent of cardiac output and regional blood flow control (📖 p64). Flow through a blood vessel is determined entirely by two factors: (1) the ***pressure difference*** between the two ends of the vessel (ΔP), and (2) the ***vascular resistance*** to flow (R). The relationship is summarized by ***Ohm's law*** (see Box 2.1).

- When blood flows through a smooth walled vessel, it flows in concentric streamlines, with the centermost stream moving fastest and streams closest to the vessel wall moving slowest: this is ***laminar flow***.
- ***Poiseulle's law***, which integrates laminar flow theory, states that flow is proportional to the 4th power of the radius of the vessel: i.e., a slight increase in vessel diameter causes flow to increase by a large amount: any decrease in vessel size causes large reductions in flow.
- Poiseulle's law also states viscosity of blood is inversely proportional to flow. ***Viscosity*** is mainly determined by hematocrit: the higher the Hb, the higher the viscosity and the slower the flow.

Box 2.1 Pressure, flow, and resistance

- Ohm's law: $Q = \Delta P/R$
- Poiseulle's law: $Q = \pi \Delta P r^4/8\eta l$

ΔP = pressure difference across two ends of a vessel, R = resistance to blood flow (Q), r= radius of the vessel, l = length of the vessel, η = viscosity of blood.

Functional parts of the circulation

The ***arteries*** transport blood under high pressure to the tissues. Inflow to the capillary beds is controlled by ***arterioles***, whose strong muscular walls can close the vessel completely or dilate by several times. The primary role of the ***capillaries*** is exchange of fluid, nutrients, electrolytes, hormones, and metabolites. ***Veins*** act as conduits back to the heart, and as a reservoir: they can expand or contract their volume as necessary.

Physiology of the coronary circulation

Normal coronary blood flow is about 225mL/min, or 5% of the cardiac output. During exertion the cardiac output increases up to six-fold, against an increased afterload: cardiac work is increased up to eight-fold. Coronary blood flow must increase to match as oxygen extraction is already near maximal: this is why flow limiting stenoses cause ischemia. The bulk of coronary flow occurs during diastole, because of the high interventricular wall pressures generated during systole. A tissue pressure gradient occurs across the myocardium: ***subendocardial*** pressures approach interventricular pressures: ***epicardial*** muscle pressures are closer to atmospheric pressures. The subendocardial blood supply is most likely to be compromised when coronary blood flow is restricted. A dense subendocardial plexus compensates for this.

Systemic circulation

The overall flow in the circulation (***cardiac output***) is about 5L/min at rest in an adult. Calculating and manipulating the cardiac output is described on 📖 p131, p184. Fig. 2.12 shows arterial traces and blood volumes in the various parts of the circulation. Although the pressure in the veins is shown as near 0mmHg, the weight of a column of blood that exists inside the veins when a person is standing or sitting still (***hydrostatic pressure***) means that venous pressure in the lower leg may be as much as 90mmHg. Arterial pressures are similarly raised. Pressures in the capillaries rise causing interstitial edema: as much as 15–20% of circulating volume may be lost from the circulation in the first 15min of standing still. The arrangement of venous valves described on 📖 p55 means that muscle contraction propels blood proximally, reducing venous pressure distally behind the valve on muscle relaxation.

Microcirculation

The capillary beds do not obey the same laws of flow and resistance just described. Once vessels are <1.5mm in diameter red cells tend to ***rouleaux*** (line up with each other and move together as a cylinder). The internal viscous resistance of blood is almost eliminated: this is called the ***Fahraeus–Lindqvist*** effect. This compensates for the much slower velocity of blood (1mm/s) in the low-pressure system. Blood flows intermittently in capillaries as a result of vasomotion, intermittent contraction of ***precapillary sphincters***. This is controlled by local oxygen demand: high oxygen demand means greater duration of flow periods. ***Diffusion*** is the most important method of transfer of solutes between the interstitium and the capillaries. The rate of diffusion is proportional to the concentration difference: solutes diffuse from areas of high concentration to areas of low concentration. Four primary forces determine osmotic movement of water and solutes: the ***capillary pressure*** and the ***interstitial colloid oncotic pressure*** which tend to move fluid out of the capillary, and the ***plasma colloid oncotic pressure*** and the ***interstitial fluid pressure*** which tend to move fluid back into the capillary. Colloid osmotic pressure is caused by negatively charged proteins, which are the only dissolved substances that do not diffuse through the capillary membrane, and associated Na^+ ions.

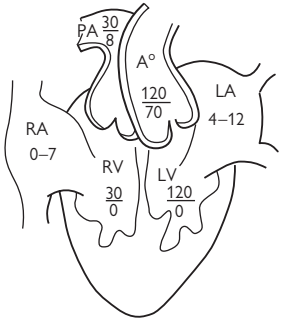


Fig. 2.12 Normal blood pressures (systolic/diastolic (mmHg)) in the cardiac chambers and great vessels. RA (right atrium), RV (right ventricle), PA (pulmonary artery), LA (left atrium), LV (left ventricle), Ao (aorta).

Regulation of blood pressure

Overview of factors contributing to blood pressure

Four main physiological mechanisms control normal arterial pressure:

- Autonomic nervous system.
- Capillary shift.
- Hormonal responses.
- Renal regulation of fluid and electrolyte balance.

Autonomic nervous system

This is the most rapid response mechanism. Continuous information is received in the brainstem vasomotor centre from central baroreceptors in the carotid sinus and aortic arch. Decreased arterial pressure activates the sympathetic nervous system resulting in increased cardiac contractility (β-adrenergic receptors), and peripheral arterial and venous vasoconstriction (α-adrenergic receptors). Increased heart rate results from inhibition of parasympathetic control.

Capillary shift

Movement of fluid between the vasculature and intersitium is controlled by hydrostatic and oncotic capillary and interstitial pressures (📖 p63). Decreased hydrostatic pressure results in fluid moving from the interstitium across the endothelium into the vasculature increasing blood volume and BP.

Hormonal mechanisms

There are two main hormonal systems, which act on different targets both resulting in rapid response to changes in arterial pressure.

- The adrenal medulla secretes endogenous catecholamines (epinephrine and norepinephrine) in response to sympathetic activation, rapidly increasing cardiac output through increased heart rate and contractility
- Decreased renal blood flow results in increased production of renin and angiotensin which is converted in the lungs to angiotensin II, a potent vasoconstrictor that stimulates production of aldosterone from the adrenal cortex which acts to decrease glomerular permeability and urinary fluid and electrolyte loss.

Renal regulation of fluid and electrolyte balance

- Renal regulation through excretion of sodium chloride (which dictates sodium balance, extracellular fluid volume, and blood volume) is the most important mechanism for long-term control of BP.
- Decreased extracellular fluid (ECF) NaCl concentrations leads to inhibition of ADH secretion by the hypothalamus, producing a diuresis, whereas increased NaCl promotes production of concentrated urine and free water reabsorption through increased circulating AHD.

Long-term sequelae

Chronic increases in the total quantity of NaCl (increased dietary intake, or decreased renal excretion) leads to a chronic increase in ECF volume, including blood volume. This may lead to edema and chronic hypertension.

Response to hypovolemia

Hypovolemia leads to an acute reduction in preload, and hence stroke volume and cardiac output. Reduced cardiac output leads to:

- Reduced carotid and aortic baroreceptor stretch, leading to inhibition of parasympathetic output, and increased sympathetic activity.
- Net result is immediate vasoconstriction, venoconstriction, increased heart rate, contractility, and increased cardiac output.
- Fluid leaves peripheral capacitance vessels and the extravascular space and shifts into the arterial tree, with blood supply to brain, heart, and kidneys maximized by autoregulation.
- Vasoconstriction is augmented by release of angiotensin II through the renin–angiotensin system and ADH. Aldosterone and ADH also lead increased salt and water reabsorption.
- As a result, blood pressure to key organs is maintained through tachycardia, vasoconstriction, venoconstriction, and decreased urine output: this is compensated shock. The patient with decompensated shock shows evidence of end-organ hypoperfusion, e.g., confusion, anuria.

Response to trauma (Fig. 2.13)

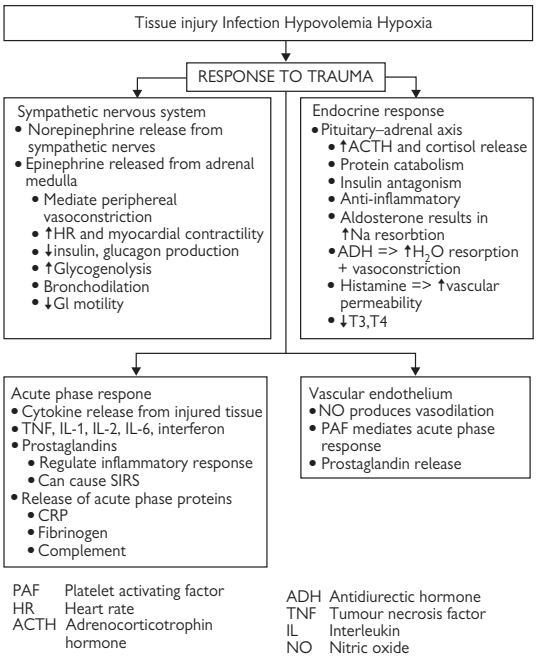


Fig. 2.13 The response to trauma.

Lung and esophageal physiology

Respiratory mechanics

Lung compliance

Lung compliance is change in lung volume per change in pressure. It may be a dynamic or static measurement and measures the distensibility of the lung parenchyma. It is directly related to lung volume, decreasing as lung volume increases. This is in contrast to the chest wall compliance, which increases with increasing lung volume. Diseases which decrease the elastic recoil of the lung increase compliance and those that increase it decrease compliance.

Surfactant

A mixture of phospholipids and protein, it is produced by type II pneumocytes. It is vital for normal lung function. Its reduces the surface tension at the alveolar–air interface. Intra-alveolar pressure may be measured by:

$$P = 2T/R$$

where P = pressure, T = tension, R = alveolar radius.

It is therefore obvious that as the alveolar radius decreases, the intra-alveolar tension increases. Surfactant reduces the surface tension proportionally to alveolar size, improving lung stability. It also increases pulmonary compliance, hence decreasing the work of breathing.

Control of ventilation

This is performed by a variety of receptors, both central and peripheral.

- ***Peripheral chemoreceptors:*** situated in the aortic and carotid bodies. They are very sensitive to changes in PCO_2. They also respond to hypoxia.
- ***Central chemoreceptors:*** situated in the ventrolateral aspect of the medulla. They respond to changes in extracellular or cerebrospinal fluid (CSF) H ion concentration.
- ***Intrapulmonary receptors:*** Herring–Breuer reflex via stretch receptors, intra-epithelial receptors responding to irritants and J receptors in the pulmonary capillary wall responding to changes in interstitial fluid volume.

Dead space

This is the volume of the respiratory system which is ventilated but does not participate in gas exchange. It includes the anatomical dead space—nasal passages, pharynx, trachea, airways up to the terminal bronchioles, and the physiological dead space—the part of tidal volume not participating in gas exchange.

$$VD/VT = (P_ACO_2 - P_ECO_2)/P_ACO_2$$

where VT = tidal volume, VD = dead space volume, P_ACO_2 = alveolar partial pressure of CO_2, P_ECO_2 = expired partial pressure of CO_2.

Diffusion capacity

This is a measure of the efficiency of diffusion. It is limited by three steps:

- The rate of oxygen passage across the alveolar–capillary membrane.

- The rate of RBC oxygen and hemoglobin combining.
- The difference in PaO_2 between the alveolus and the erythrocyte.

Flow–volume loops (Table 2.2)

Table 2.2 Diagnosis of lung disease (see also Table 13.1)

	Obstructive lung disease	Restrictive lung disease
FEV_1	↓	↓
FVC	Preserved	↓
FEV_1/FVC	↓	↑

Normal esophageal function

Upper esophageal sphincter

Separates the pharynx and the esophagus and prevents regurgitation. Formed from the cricopharyngeus muscle.

Esophageal body

Longitudinal and circular muscle. Circular muscle involved in peristalsis. Commences upon swallowing after which time it is under involuntary control. Peristaltic wave travels at 2–5cm/s along the esophageal body in a cranial to caudal direction, hence propelling esophageal contents towards the lower esophageal sphincter and the stomach.

Lower esophageal sphincter

Has a basal tone to prevent continuous reflux of gastric contents into the esophagus. Relaxes to allow esophageal contents to enter the stomach. Both a physiological and an anatomic sphincter. Relaxation occurs with esophageal distension, swallowing and with distension of the gastric fundus (see 📖 p50, and Fig. 13.4 on 📖 p659 for normal barriers to reflux).

Esophageal pH studies

24h monitoring allows quantitative measurements of acid reflux. It is the standard test when diagnosing gastroesophageal reflux disease (see 📖 p778) and is also useful when investigating patients with recurrent symptoms after surgery for reflux and those with symptoms of reflux following surgery for achalasia. Normal esophageal pH is 4–7. A reading below 4 is registered as an episode of reflux, but cannot identify the cause for the reflux.

- All anti-reflux and antacid medications are stopped prior to the study.
- A probe is placed 5cm above the upper border of the lower esophageal sphincter.
- Patients are advised to be active and to keep accurate diaries documenting meal times, symptoms, and when supine or erect.

Renal homeostasis

Key facts

By controlling solute and water excretion, and hormone production, the kidneys have a key role in control of BP, volume and osmolality, acid–base balance, electrolyte control, and excretion of toxins.

Renal function

GFR is decreased by low renal blood flow, high osmotic plasma pressures, and low hydrostatic capillary pressures, which are determined by afferent and efferent arteriolar resistance. Sympathetic drive reduces renal blood flow and GFR. Creatinine excretion is used to measure GFR.

Clearance and glomerular filtration rate (GFR)

Clearance (Cl) is the volume of plasma completely cleared of solute by the kidney (ml/min) and is the main measure of kidney function.
$Cl = (U/P) \times V$

GFR = clearance, for any substance not metabolized, actively excreted or reabsorbed in the tubules = 125mL/min (in normal adults); U = concentration in urine; P = concentration in plasma of solute; V = urine flow rate.

Body fluid compartments

(See Table 2.3.) In a typical 70kg, 1.80m male:

- Total body water is 42L (60% of body mass).
- Intracellular fluid (ICF) is 27L (40%).
- Extracellular fluid (ECF) is 15L (60%).
- Of the ECF, total blood volume is 5L.
- 2L is contributed by red cell volume.
- Daily intake averages at 2L.
- Insensible loss from the respiratory tract and skin is about 700mL. Daily requirements of solutes are: 160mmol Na^+, 100mmol K^+, 950mmol Ca^{2+}, and 300mmol Mg^{2+}.

Table 2.3 Typical contents of body fluids (mmol/L)

Solute	Plasma	ECF	ICF	Lymph
Na^+	142	145	12	140
K^+	4.3	4.4	150	
Ca^{2+}	2.5	2.4	4.0	
Mg^{2+}	1.1	1.1	34	
Cl^-	104	117	4.0	
PO_4^{2-}	2.0	2.0	40	
$Prot^-$	14	0	54	
HCO_3^-	24	27	12	

Renal tubular function

GFR is autoregulated closely: even a 5% error in matching GFR to body requirements results in rapid accumulation of waste products or excess loss of water and solutes into the urine. Two feedback mechanisms accomplish autoregulation at each glomerulus: (1) an afferent arteriolar vasodilator, (2) an efferent arteriolar vasoconstrictor. These control GFR by controlling blood flow into and out of the individual glomerulus. In this way renal blood flow is kept constant through a wide range of systemic pressures. Urine is formed from the glomerular filtrate by a process of active reabsorbtion, active secretion, changing membrane permeability to water and a countercurrent multiplier (repetitive reabsorption of NaCl by the thick ascending loop of Henle together with continual inflow of new NaCl from the adjacent proximal tubule into the loop of Henle, 'multiplying' the concentration of NaCl).

- Normal pH is 7.35–7.45.
- Normal H^+ concentration is 35–45nmol/L.
- pH is the negative log of H^+ concentration.
- Base excess is the amount of base required to restore 1L of blood to pH 7.0 at a pCO_2 of 5.3kPa.
- Anion gap is the amount of anions other than Cl^- and HCO_3^- (effectively $H_2PO_4^-$ and SO_4^-) needed to neutralize measured Na^+ ions.

Acid–base balance

The kidney controls blood pH by secreting alkaline or acidic urine, by balancing the amount of HCO_3^- reabsorbed and the amount of H^+ secreted. There are three major buffer systems, which act across organ systems: (1) the bicarbonate, (2) the phosphate, and (3) the protein buffer system. The extracellular protein buffers can correct acid–base changes within seconds, the respiratory system can correct changes within minutes, but renal mechanisms (primarily bicarbonate) take hours to days. This longevity makes it more powerful than the other two systems.

Blood

Blood groups

There are 30 commonly occurring and hundreds of rare antigens present on human blood cells. Most do not elicit strong antigen–antibody reactions. The antibodies to two related antigens, ***type A*** and ***type B***, almost always occur in the plasma of people who do not have the antigens on their red cells. These antibodies bind strongly with the red cell antigens to cause agglutination, and form the basis for blood typing. The ***rhesus*** system is the other important system, but in order for a person to develop rhesus antibodies they must first be exposed to the rhesus antigen, which happens by blood transfusion, or by a Rh −ve mother giving birth to a Rh +ve baby. Of the six rhesus antigens, rhesus D is the most antigenic in the US:

- 37% are blood group O (have neither A nor B antigens on red blood cells, but have anti-A and anti-B antibodies in their plasma).
- 36% are group A (A antigen, anti-B antibodies).
- 9% are group B (B antigen, anti-A antibodies).
- 3% are AB (A and B antigens, no A or B antibodies).
- O Rh –ve patients are ***universal donors*** (their blood can be given to anyone) because their blood cells carry antigens from neither system.
- AB +ve patients are ***universal recipients*** as their blood contains antibodies to neither system.

Cross-matching blood

- ***Type and screen*** (or blood grouping) involves adding A and B agglutinins to donated blood to determine blood type.
- ***Cross-matching*** involves mixing donated blood with the intended recipient blood, to assess compatibility. Cross-matching takes 20min. If no cross-matched blood is available, O –ve blood may be given in an emergency, but rare antigen systems may occasionally cause a transfusion reaction (📖 p148). Immunocompromised patients may experience a form of ***graft-vs-host disease*** after transfusion, when donor T cells attack host antigens.

Tissue typing

For lung transplants ***major histompatibility complex (MHC)*** antigens, also known as ***human leucocyte antigens (HLA)***, must be typed in addition to the ***ABO system*** if rejection is to be avoided. Chromosome 6 contains the MHC complex that codes for expression of MHC antigens, which originally evolved to enable to the body to detect and combat infecting organisms. Most heart transplants do not need an HLA match (📖 p507).

- MHC class I (A, B, and C) antigens are expressed on the surface of all nucleated cells and platelets.
- MHC class II (DP, DQ, and DR) antigens are expressed on B-lymphocytes, macrophages, monocytes and dendritic cells.
- Each antigen is highly polymorphic: e.g., there are >70 different versions, or alleles, of the DR locus; while there are six ABO phenotypes, there are over 10^{12} MHC phenotypes.
- Tissue typing can be done by using panels of antisera, in a manner similar to blood typing, or by PCR, which allows automation.

Hemostasis and fibrinolysis

There are four phases of hemostasis:

- ***A vascular phase:*** local BP and flow are reduced by vasospasm in response to direct mechanical and local humoral effects, and local edema and hematoma formation reduces transmural pressure.
- ***A platelet phase:*** adherence of platelets to damaged endothelium activated by ADP and collagen, and binding von Willebrand factor. Platelet aggregation is mediated by membrane glycoprotein IIb/IIIa and fibrinogen, 5HT, thromboxane A2. Activated platelets release thromboxane A2 and platelet factor.
- ***A clotting phase:*** the intrinsic and extrinsic pathways happen in parallel *in vivo*.
- ***Monocyte activation:*** they express tissue factor and factor V.

Fibrinolysis occurs at the same time as clot formation, to limit the process locally (see Fig. 2.14). Fibrinolysis depends on four main molecules. (1) ***Plasmin***, a serine protease which is produced by the action of thrombin on plasminogen and attacks unstable bonds between fibrin molecules to generate fibrin degradation products. (2) ***Antithrombin III*** which binds thrombin, XIIa, IXa, and XI to deactivate them. (3) ***Proteins C and S*** which prevent thrombin generation by binding factors Va and VIIIa. (4) ***Tissue factor pathway inhibitor***, produced by platelets inhibits factors Xa and VIIa.

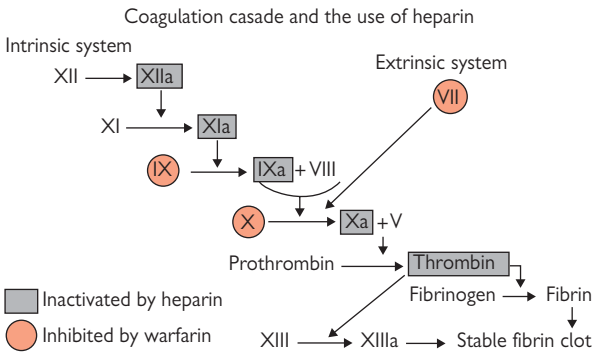


Fig. 2.14 Diagram of the clotting and fibrinolysis cascades.

Ischemic heart disease

Atherosclerosis

Key facts

- Atherosclerosis is a degenerative disease of large and medium-sized arteries characterized by lipid deposition and fibrosis.
- There are ***three stages*** of atheromatous lesion; ***fatty streaks*** are linear lesions on the artery lumen, composed of lipid-filled macrophages, and which progress to ***fibrolipid plaques*** (unstable plaques), and finally ***complex lesions*** (stable plaques).
- There are multiple lesions throughout the whole vascular tree at all stages of the disease process: the flow-limiting ***culprit lesion*** responsible for stable angina is usually a stable plaque, whereas the unstable plaques responsible for acute MI are fibrolipid plaques, and are often not due to the flow-limiting lesions visible on coronary angiography.
- Reversible risk factors: smoking, hypercholesterolemia, obesity, hypertension. Irreversible risk factors: diabetes mellitus, male sex, age, family history.

Pathological features

- In sites at risk of atherosclerosis (sites of vessel bifurcation, turbulent flow, post-stenotic areas, areas denuded of endothelial cells) lipid-laden macrophages enter the vessel wall via gaps between endothelial cells.
- A fibrolipid plaque contains a mix of macrophages and smooth muscle cells which migrate into the plaque, capped by fibrous tissue (Fig. 2.15).
- Growth factors, particularly platelet-derived growth factor (PDGF), stimulate the proliferation of intimal smooth muscle cells and the synthesis of collagen, elastin, and mucopolysaccharide.
- Lipid accumulates within the plaque extracellularly, and in the myocytes, ultimately producing ***foam cells***.
- Cell death eventually ensues with the release of intracellular lipids, calcification, and a chronic inflammatory reaction.
- High levels of circulating ***LDL-cholesterol*** are thought to lead to atherosclerosis by damaging endothelium both directly by increasing membrane viscosity, indirectly through ***free radical*** formation, and by inducing secretion of PDGF.
- In larger vessels such as the aorta, atherosclerotic plaques may release ***atheroemboli*** and ***mural thrombus***, or impinge on the vessel media causing tissue atrophy resulting in ***aneurysm*** formation or ***dissection***.
- ***Acute MI*** is caused by three processes in coronary vessels: progressive atherosclerosis, disruption of ***unstable plaque*** with ***acute thrombosis***, and acute hemorrhage into the intima around the plaque.

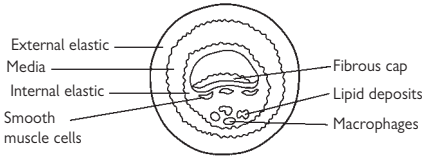


Fig. 2.15 Cross-section through an atherosclerotic plaque.

Ischemia and infarction

Key facts

- *Ischemia* is cell damage caused by O_2 supply–demand mismatch.
- *Infarction* is cell death caused by O_2 supply–demand mismatch.
- Normal O_2 supply to the contracting LV is 8mL per 100g muscle per minute. Imbalance in O_2 supply and demand results in ischemia—changes are most acute in the subendocardium as flow mostly occurs in diastole (📖 p62). Supply–demand mismatch is caused by:
  - Vascular narrowing (atherosclerosis, thrombus, embolus, spasm).
  - Global hypoperfusion (shock, cardiopulmonary bypass).
  - Hypoxemia (anemia, hypoxia).
  - Vascular compression (increased LVEDP in CHF, distension on bypass).
  - Increased myocardial O_2 demand (exercise, pregnancy, hyperthyroidism, tachycardia, ventricular hypertrophy, ventricular distension).

Pathological features

MIs occur in the right coronary artery territory in 30%, left anterior descending in 50% of cases, and circumflex in 20% of cases.

Changes within seconds of ischemia (reversible)

- Switch to anerobic glycolysis from oxidative metabolism.
- Decrease in high-energy phosphates (creatinine and ATP)—this can impair recovery after reperfusion, hence need for rapid electromechanical arrest after cross-clamping to avoid depleting myocardial ATP.
- Accumulation of lactic acid and rise in myocardial pH.

Changes within minutes (reversible if oxygen supply restored)

- Decrease in contractility.
- Glycogen stores depleted.
- Cell and mitochondrial swelling.

Changes after 30 minutes of ischemia (irreversible)

- Structural defects appear in the sarcolemma.
- Myocyte death.

Changes after 3 hours

- Coagulation necrosis.
- Changes are only visible macroscopically at 24 hours: muscle appears pale and edematous.

Changes after 3 days

- Inflammatory exudates initially with polymorphonuclear leukocytes.
- Necrotic tissue removed by macrophages.
- Fibroblast infiltration beginning the process of scar formation.
- Macroscopically the infarcted area appears yellow and rubbery with a hemorrhagic border.
- Aneurysm formation and free wall rupture may occur while myocardial wall is weak, before formation of scar tissue.

Changes after 1 week

- Neorevascularization at margins of preserved tissue.
- Scar maturation normally complete by 6 weeks: tough, white area.

Ischemia–reperfusion injury

Key facts

Ischemia–reperfusion injury occurs after angina, spontaneous lysis of coronary thrombi, relief of coronary spasm, thrombolysis, PCI, off-pump coronary artery bypass (OPBCAB), and surgery with electromechanical arrest.

Myocardial stunning

Myocardial stunning

The mechanical dysfunction that persists after reperfusion in the absence of irreversible damage. It is usually a relatively mild and fully reversible injury that is distinct from myocardial infarction, but it contributes to the morbidity and mortality of the ischemic injuries listed in this chapter. The degree of myocardial stunning is proportional to the antecedent ischemia, suggesting that ischemic changes initiate and potentiate reperfusion injury. There are three main mechanisms of injury:

Oxygen free radicals

Generated in the stunned myocardium causing damage during the initial moments of reperfusion. They are produced by several pathways which include increased xanthine oxidase activity, activation of neutrophils, deranged intramitochondrial electron transport systems and auto-oxidative processes. The free radicals react with proteins and fatty acids. Oxygen radical mediated damage of the sarcoplasmic reticulum, sarcolemma, extracellular collagen matrix, and intracellular contractile proteins results in a large rise in intracellular calcium.

Rise in cytosolic free calcium

The rise in cytosolic free calcium activates degradation enzymes such as the phospholipases, further damaging the same intracellular structures and amplifying the damage done by the oxygen free radicals, resulting in reduced myocardial contractility. A prolonged intracellular calcium deficit ensues which is addressed by administration of exogenous calcium.

Excitation-contraction uncoupling

The oxygen radical mediated damage of the sarcoplasmic reticulum, sarcolemma, extracellular collagen matrix and intracellular contractile proteins results in uncoupling of the excitation–contraction mechanism.

The oxygen paradox

This is an irreversible phenomenon. Reoxygenation of ischemic myocardial cells leads to rapid acceleration of the damage described in this chapter. It is characterized by cellular contracture, sudden rupture of increasingly fragile cell membranes, and the release of cell contents as a result of abrupt changes in pH and tissue osmolality, and it seems to occur in the no man's land between reversible and irreversible ischemia. It is an energy-dependent process that does not occur when the cell's ability to generate oxidative energy is abolished, and it only occurs in myocytes. The process is triggered by the re-supply of energy from oxidative phosphorylation to myofibrils, which due to elevated cytosolic calcium are highly activated: hypercontracture and cell death is the result.

Neutrophils and ischemia

Neutrophils accumulate rapidly. In response to acute MI the neutrophil response is apparent at 12–24h, peaking at 3–4 days after permanent coronary artery occlusion. In reperfusion injury neutrophil infiltration begins earlier, is accelerated by reperfusion, and is proportional to the preceding ischemia. Neutrophil depletion or inactivation during reperfusion results in smaller infarct size. Neutrophils probably exacerbate reperfusion injury in three ways:

- Plugging capillaries reducing collateral flow.
- Releasing vasoconstrictors.
- Acting as a source of oxygen free radicals produced by the NADPH-oxidase reaction. 70% of the oxygen used by activated neutrophils is converted to superoxide. This is the basis of their attack on bacteria and damaged cells, but in the myocardium it causes dysfunction.

Quantifying ischemia

- Methods of quantifying ischemia in real-time include:
- Measurement of high-energy phosphates such as ATP, and their breakdown products (purines and pyrimidines) in coronary sinus outflow.
- Measuring levels of lactate in coronary sinus blood.
- Measuring intraoperative myocardial pH with glass electrodes.

Ischemic preconditioning

One or more brief periods of myocardial ischemia followed by reperfusion increases the ability of the myocardium to withstand longer periods of ischemia. Myocardial ATP declines during the first short ischemic periods, but not during subsequent prolonged ischemia. The number of myocytes that die during a period of prolonged ischemia are reduced by up to 75% if the ischemic period is preceded by a shorter period of ischemia. This is because the metabolic changes described in this chapter, when by triggered by a very brief period of ischemia become adaptive rather than maladaptive.

Hibernation

Normally 8mL of oxygen per 100g of muscle per minute is delivered to the contracting left ventricle. Cellular viability is compromised when this falls below 1.5mL oxygen per 100g per minute. With repeated episodes of stunning (such as occurs in stable angina due to coronary artery disease) the metabolic processes of the heart remain intact, but the regional contractility may be reduced. This is ***hibernation***: with reperfusion hibernating myocardium may resume normal contraction. Detecting hibernating myocardium is described on 📖 p28.

Pathology of heart failure

Key facts

Heart failure is an inability of the heart to produce an adequate cardiac output despite normal filling pressures. It is characterized by numerous multisystem maladaptive changes. The reduction in cardiac function that underlies these changes may be thought of as ***systolic*** (a decrease in contractility) and ***diastolic*** (a decrease in compliance, and hence decreased filling and stroke volume).

Classification

Heart failure may be classified in several different ways:

- Acute and chronic.
- Left, right, and biventricular or congestive.
- High and low output.
- Systolic and diastolic.
- Compensated and decompensated.

40% of patients with heart failure have near normal systolic function: their failure is primarily diastolic.

Etiology

There are many causes of cardiac failure (see Box 2.2), but in the West coronary artery disease and/or hypertension feature in >90% of cases. Severe forms of these disease processes all end in a common pathway of systolic or diastolic dysfunction that results in heart failure.

Box 2.2 Causes of heart failure

- Hypertension.
- Ischemic heart disease.
- Valvular heart disease.
- Toxic: alcohol, cocaine, β-blockers, Ca channel blockers, adriamycin, doxorubicin.
- Viral myocarditis: Coxsackie, HIV.
- The cardiomyopathies: hypertrophic, restrictive, dilated.
- Primary pulmonary hypertension.
- Arrhythmias: AF, sick sinus syndrome.
- Infiltrative: amyloidosis, sarcoidosis.
- Pericardial disease: restrictive pericarditis, pericardial effusion.
- Metabolic: thyrotoxicosis.
- Nutritional: Beri-beri, pellagra.
- High output: anemia, Paget's disease, pregnancy.
- Neuromuscular: Duchenne muscular dystrophy.

Pathogenesis

- The wide ranging changes that occur in heart failure are divided into cardiac and non-cardiac.

Cardiac changes in heart failure

Myocyte

Abnormal excitation–contraction, β-adrenergic downregulation.

Myocardium

Remodelling: regional hypertrophy (eccentric in volume overload, and concentric in pressure overload), thinning and dilatation of infarct zones, increased sphericity, necrosis, fibrosis.

Coronary arteries

Extrinsic compression, inflammation.

Functional changes

MR, ischemia, hibernation, arrhythmias, failure of coordination of right and left ventricular contraction due to heart block.

Non-cardiac changes in heart failure

The fall in cardiac output leads to activation of several neurohumoral systems designed to maintain cardiac output.

- Renin–angiotensin–aldosterone system (RAAS).
- Sympathetic nervous system.
- Brain and atrial natriuretic peptide.

RAAS activation leads to increased circulating renin, plasma angiotensin II, and aldosterone. ***Angiotensin II*** is a potent vasoconstrictor of renal efferent arterioles and systemic circulation where it stimulates release of ***norepinephrine*** from sympathetic nerve endings, stimulates release of ***aldosterone***, and inhibits vagal tone. The net result is ***sodium*** and ***water retention***, and potassium excretion.

Sympathetic activation occurs early in heart failure via low- and high-pressure baroceptors, providing inotropic support and chronotropic drive to maintain cardiac output. Sustained sympathetic activation activates the RAAS, increasing arterial and venous tone, preload and afterload, and leading to progressive retention of sodium and water. Sympathetic overdrive results in ventricular hypertrophy and focal myocardial necrosis.

Increase in circulating ***atrial*** and ***brain natuiretic peptide*** in response to volume expansion leads to ***natruriesis*** and vasodilation: antagonizing the effects of aldosterone. ***ADH*** may contribute to hyponatremia. Endothelin, secreted by endothelial cells, acts to conserve sodium and stimulate vasoconstriction.

Valvular heart disease

The pathophysiology of valvular heart disease is described in more detail in 📖 chapter 8. The following section concentrates on the morphology.

Key facts

The content and arrangement of valve fibres (📖 p44) is designed to give maximum strength and flexibility, and minimum obstruction to flow where required. Any pathological process affecting valve structure has a large impact on its function.

Aortic stenosis

Etiology

- Congenital (unicuspid rare, ***bicuspid valve*** more common around 1–2%). Bicuspid valve morphology varies—may be two equal cusps with central opening, or two unequally sized cusps with raphe in larger cusp indicating where two leaflets have fused. Bicuspid aortic valves usually functionally normal in younger patients, but leaflets may become increasingly sclerotic with age, leading to accelerated stenosis. Around 50% of bicupisd aortic valve are stenotic by age 60 years. Bicuspid valves associated with aortic root dilatation (📖 p428).
- ***Rheumatic*** (📖 pp372 and 398) due to commissural fusion, leaflet thickening.
- ***Calcific degeneration:*** commonest cause, occurring in otherwise normal valves. Rheumatic and bicuspid valves eventually calcify.
- Infective endocarditis (rare cause of AS, usually causes AI).
- Hyperlipidaemia (rare).
- Subvalvar (membrane and muscular) and supravalvular.
- Prosthesis failure (pannus, thrombosis, endocarditis, calcification).

Aortic insufficiency

Etiology

- Myxomatous degeneration (common cause, causes leaflet prolapse).
- Rheumatoid (often mixed picture with degree of AS).
- Infective endocarditis (leaflet perforation).
- Root dilatation (quite common—due to rheumatic, atherosclerotic, aneurysmal, Marfan syndrome, syphilis, ankylosing spondylitis).
- Prosthesis failure: paraprosthetic leak, leaflet perforation.

Functional classification

- Disease of the aortic root, e.g., Marfan syndrome.
- Disease of the valve leaflets, e.g., rheumatic, endocarditis.
- Lack of support of the valve commissures, e.g., dissection, trauma.

Mixed aortic valve disease

Mild AI frequently accompanies AS. The commonest causes of mixed aortic valve disease are bicuspid aortic valve, chronic rheumatic valve disease, and infective endocarditis in a stenotic valve.

Mitral stenosis

Etiology

- Chronic ***rheumatic*** heart disease (commonest cause).
- ***Congenital*** mitral stenosis (rare).
- ***Mitral annular calcification*** (MAC) (less usual cause of severe MS).

Mitral regurgitation

Pathophysiologic and functional classifications are described on 📖 p400.

Aetiology

- Degenerative mitral valve disease. Commonest cause in West. Represents spectrum from fibroelastic deficiency (small valves, single segment prolapse), to Barlow's disease (large valves, multisegment prolapse). Also called floppy valve, prolapse, myxomatous disease.
- Rheumatic heart disease.
- Infective endocarditis
- Connective tissue disorders, e.g. Marfan and Ehlers–Danlos syndromes.
- Ischemic heart disease.
- Congenital cleft valve leaflet (associated with primum ASD).
- Endomyocardial fibrosis (common in sub-Saharan Africa).
- Iatrogenic (balloon valvuloplasty of stenotic valve).
- Prosthesis failure (paraprosthetic leak, leaflet perforation).

Functional classification

Clear understanding of the functional classification (📖 p400) is required for successful repair. Essentially ***Carpentier's functional classification*** describes Type I (normal leaflet motion, MR caused by annular dilatation or leaflet perforation), Type II (excess leaflet motion, MR caused by leaflet prolapse due to chordal or papillary muscle elongation or rupture), Type IIIIa (decreased diastolic leaflet motion due to leaflet thickening) and Type IIIb (decreased systolic leaflet motion due to chordal tethering or shortening.

Mixed mitral valve disease

- The commonest cause of mixed mitral valve disease is rheumatic. Usually MS predominates.

Tricupid valve disease

TS is rare—usually rheumatoid. TR may be caused by:

- ***Functional*** secondary to mitral valve disease (commonest).
- ***Rheumatic*** heart disease.
- ***Infective endocarditis***.
- Ebstein's anomaly.
- ***Carcinoid*** syndrome (usually associated with pulmonary regurgitation).
- Endomyocardial fibrosis.
- Prolapsing cusp.

Pulmonary valve disease

Acquired pulmonary valve disease is unusual. Severe pulmonary hypertension may cause dilatation of the pulmonary valve ring causing pulmonary regurgitation.

Infective endocarditis

Key facts

- Infective endocarditis is a microbial infection of the endocardial surface of the heart.
- Incidence 1.7–6.2 cases per 100 000 person years in the West.
- Men more commonly affected than women.
- In MV prolapse the incidence is 100 per 100 000 person years.
- In iv drug users the incidence is 150–2000 per 100 000 person years.
- Prosthetic valve endocarditis accounts for up to 25% of cases in the West. The cumulative risk is about 1% at 12 months and 3% at 5 years post surgery.

Pathology

Normal cardiac endothelium is resistant to infection. Transient bacteremias occur in everyone but immune mechanisms, particularly platelet thrombocidins, play an effective role in preventing endocarditis. Damage to cardiac endothelium, as a result of trauma, turbulent flow, valvular heart disease, or atherosclerosis results in platelet and fibrin deposition. A sterile thrombotic vegetation is produced. Microbes from peripheral sites can colonize these vegetations to cause infective endocarditis, monocyte adhesion, triggering of the coagulation cascade, and clot formation. Commonest sites of infective endocarditis are:

- Predominantly left-sided (95%).
- Mitral valve (85%), aortic valve (55%), tricuspid valve (20%).
- The pulmonary valve is involved in <1% of cases.
- Jet lesions: atrial surface of the mitral valve in MR, or ventricular surface of the aortic valve in AR.

Microbiology

Staphylococcus

Staphylococci, particularly *Staphylococcus aureus*, are now the commonest cause of infective endocarditis, causing up to 40% of cases of native valve endocarditis, and up to 25% of cases of prosthetic valve endocarditis. Coagulase-negative staph cause up to 30% of cases of prosthetic valve endocarditis but <10% of cases of native valve endocarditis. *Staph. aureus* endocarditis is particularly virulent, and associated with annular and myocardial abscess formation and a higher mortality.

Streptococcus

Viridans streptococci, a group of bacteria usually from dental caries, were until recently the commonest cause of endocarditis. Bacteremias can be caused by tooth brushing and chewing, not just by dental extraction. Streptococci now account for about 60% of native valve endocarditis and up to 10% of prosthetic valve endocarditis. The common isolates are *Strep. sanguis, bovis, mutans*, and *mitis*. *Strep. bovis* is most common amongst elderly people with bowel pathology.

Enterococci

Enterococci are frequently isolated in hospital-acquired bacteremias and cause about 10% of cases of native valve endocarditis, and a similar proportion of prosthetic valve endocarditis. They are virulent.

HACEK species

HACEK stands for *Haemophilus* species, *Actinobacillus actinomycetemcomitans, Cardiobacterium hominis, Eikenella corrodens*, and *Kingella kingae*. These organisms are slow-growing oral commensals and their presence in blood cultures is almost pathognomonic of infective endocarditis. HACEK vegetations tend to be large.

Polymicrobial infective endocarditis

This is unusual, and normally associated with iv drug abuse.

Sterile endocarditis

Up to 10% of cases of proven infective endocarditis (proven by operative or postmortem cultures) yield negative blood cultures. Previous antibiotic therapy is the commonest cause. *Coxiella burnetii, Bartonella* spp., and *Chlamydia* spp. cannot be grown by conventional blood culture methods.

Clinical features

The clinical presentation is extremely varied. The Duke diagnostic criteria are listed on 📖 p418. Symptoms and signs include:

Bacteremia

Pyrexia, chills, rigors, malaise, anorexia, confusion, arthralgia.

Tissue destruction

Embolic illness, valvular incompetence, root and myocardial abscess and fistula formation, dehiscence of prosthetic valves and false aneurysm formation. Heart block results from erosion of the conducting tissues.

Emboli

Emboli occur in up to 50% of cases of infective endocarditis, commonly resulting in CVA (65% of emboli), acute peripheral limb ischemia, MI, TIA, spleen and kidney infarction, PE, and mycotic aneurysms.

Circulating immune complexes

These are manifested by splinter hemorrhages, Osler nodes, and Janeway lesions (📖 p6), vasculitic rash, Roth spots, splenomegaly, nephritis, and arthralgia.

Mortality

The mortality rate depends on the organism and the valvular pathology:

- Up to 50% for *Staph. aureus* and fungi.
- Up to 15% for viridans streptococci and *Strep. bovis*.
- Up to 25% for enterococci.
- About 10% in right-sided endocarditis in iv drug users.
- The overall mortality for native and prosthetic valve endocarditis is 25%, but late prosthetic endocarditis has a mortality of up to 60%.

The management of infective endocarditis is described on 📖 p418.

Pericardial pathology

Etiology

Pericardial disease can be classified as follows:

- Acute and chronic pericarditis.
- Chronic pericarditis may have constrictive component. Severe constrictive pericarditis may be an indications for surgery (📖 p488) but should not be confused with restrictive cardmyopathy (Table 2.4) which may present similarly, but requires different management.
- Infective: viral, bacterial, toxoplasmosis, amebiasis, TB, ecchinococcal.
- Inflammatory: Dressler's, post cardiotomy.
- Connective tissue disorders: RA, SLE, PAN, rheumatic fever, sclerosis.
- Systemic disease: uraemia, hypothyroidism.
- Neoplastic: primary or metastatic (📖 p84).
- Effusion and hemorrhage: trauma, aortic dissection.
- Physical agents: radiotherapy, blunt trauma.

Pathophysiology

Normally the pericardial volume is 10% greater than that of the heart. Slow accumulation of fluid in the pericardial space can be accommodated by changes in the pericardium over time, but acute increases in the amount of fluid by even small amounts, and constrictive pericardial disease processes, produce pericardial compression (cardiac tamponade).

Pulsus paradoxus

This is an exaggeration of a normal physiological phenomenon. In inspiration systemic systolic pressures fall by about 4mmHg. In cardiac tamponade this fall is over 10mmHg. This is because inspiration requires the generation of negative intrathoracic pressures, effectively drawing increased blood volumes into the chest and right heart. The interventricular septum is displaced to the left, reducing left ventricular filling. This translates into reduced stroke volume and reduced systemic pressures. The second impact of reduced intrathoracic pressures is a direct effect on the left ventricle: the force that the left ventricle needs to exert relative to the positive pressure in extrathoracic arteries is greater, and the left ventricle empties incompletely during full inspiration. In cardiac tamponade external compression further reduces ventricular filling (LVEDV decreases by up to 30%) exaggerating the decrease in systolic pressure in inspiration.

Equalization of pressures

- LVEDP and RVEDP are within 5mmHg of each other.

Elevation of mean atrial pressures

- RAP and LAP >10mmHg are diagnostic requirements.

Square root sign

A dip plateau pattern in the ventricular filling pressure curve reflects normal ventricular filling that suddenly reaches the elastic limits of constricted pericardium, as opposed to gradually reaching the limit (Fig. 2.16).

Prominent y descent

- In the jugular venous pressure trace x and y descents are prominent.

Elevated RVEDP

- This is typically more than a third of the RV systolic pressure.

Table 2.4 Differences between restrictive cardiomyopathy and constrictive pericarditis

	Restrictive cardiomyopathy	Constrictive pericarditis
Bedside	Usually no history of surgery Pansysolic murmurs from TR and MR common, with ↑apex beat	History of previous cardiac surgery common Murmurs uncommon.
	Kussmaul sign unusual	Kussmaul sign present
Echo	↑Wall thickness	Normal or reduced wall thickness
	No respiratory variation in transmitral/tricuspid flows	>25% respiratory variation in transmitral/tricuspid flows
	No septal bounce	Septal bounce
Cardiac cath	Square root sign present	
	Elevated RA pressures	
	Pulsus paradoxus	
	LVEDP >RVEDP	Equalization of EDPs and RAP
	Concordance in RV and LV pressure changes with respiration	RV and LV pressure discordance
	Slow early diastolic filling	Rapid early diastolic filling
CT ? MRI	Normal pericardium	Thickened pericardium, sometimes calcified
Endocardial biopsy	Normal pericardium Amyloid on endomyocardial biopsy	Inflammatory, calcified (rare to require pericardial biopsy)

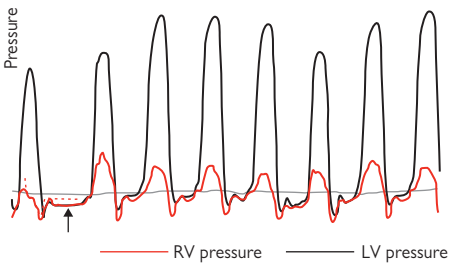


Fig. 2.16 Simultaneous pressure recordings of RV and LV pressures showing equalization of filling pressures (RVEDP and LVEDP), concordance of respiratory variation in RV and LV pressures, and dip and plateau or 'square root' sign (arrowed).

Cardiac neoplasms

Classification of cardiac neoplasms

Primary

- Benign: myxoma, lipoma, fibroblastoma of valves, rhabdomyoma, mesothelioma of the AV node, hemangioma, teratoma, pheochromocytoma.
- Malignant: angiosarcoma, rhabdomyosarcoma, malignant mesothelioma, fibrosarcoma.

Secondary

- Metastatic: melanoma, sarcoma, bronchiogenic carcinoma, adenocarcinomas of prostate and bowel.
- Direct spread: from lung, breast, esophagus, thymus, hepatic, adrenal, uterine, and renal cell carcinomas.

Myxoma

Definition

Myxoma is a neoplasm of endocardial origin, derived from either subendocardial pluripotential mesenchymal cells or endocardial nerve cells.

Etiology

- Incidence 1:100,000, Commoner in women.
- Comprises 50% of benign cardiac tumors in adults (15% in children).
- Peak incidence 3rd–6th decades.
- 5% show a familiar pattern of autosomal dominant inheritance.
- Myxomas have developed after cardiac trauma.

Pathological features

Myxomas are usually smooth, ovoid, polypoid, and mobile. Less common forms are sessile, villous, and papillary. Most are firm, but some may be gelatinous and friable. Most contain areas of hemorrhage, cyst formation or necrosis. The average size is about 5cm, but many are larger. Histology reveals polygonal cells and capillaries within a mucopolysaccharide matrix. Myxomas tend to grow into the overlying cardiac cavity rather than the surrounding myocardium. They occur in all chambers:

- 75% are LA, 20% are RA, 8% are ventricular.
- Multicentric tumors are commoner in familial disease.
- Atrial tumors commonly arise from the borders of the fossa ovalis.
- RA tumors tend to be broader based and calcified.

Clinical features

- ***Embolism:*** 30–40% of patients, mostly systemic but also cerebral (including retinal). PE is rare.
- ***Intracardiac obstruction:*** decreased RV or LV filling (lie patient down to dislodge tumor from outflow tract), pulmonary edema, RVF, syncope, sudden death, MR 'wrecking ball', patent foramen ovale (PFO) and shunting cyanosis. Removal described on p473.

- ***Constitutional:*** fatigue, fever, rash, pyrexia, arthralgia, myalgia, decreased weight, decreased Hb, increased CRP and ESR.
- ***Infective:*** infected myxoma and septic emboli.

Other primary benign tumors

Myxomas account for 40% of benign tumors, lipomas, papillary fibroelastomas, and rhabdomyomas account for another 40% of benign tumors, and a number of rarer tumors account for the remaining 20%.

Lipomas

Lipomas are encapsulated tumors of fat cells occurring predominantly in the pericardium, subendocardium and interatrial septum. RA and LV are the sites most commonly affected. The tumors are slow growing and usually asymptomatic. Subendocardial tumors may produce chamber obstruction, most commonly of the RA and LV. Non-encapsulated lipomatous hypertrophy of the interatrial septum is more common than cardiac lipoma. It is encountered in elderly, obese, and female patients. It is not an indication for surgery.

Papillary fibroelastomas

These tumors arise from the cardiac valves or endocardium. All four valves are affected equally frequently, and the villous tumors may occasionally produce flow obstruction and more commonly emboli. These tumors should be resected to prevent embolic complications.

Rhabdomyomas

Rhabdomyomas are probably myocardial hamartomas. They classically occur in children. 90% are multiple, and most are ventricular. They tend to present with flow obstruction, and early operation is recommended.

Primary malignant tumors

These tumors are rare, and largely incurable. They usually arise in adults over the age of 40. Angiosarcomas are 2–3 times commoner in men and 80% arise from the RA. They are bulky tumors which aggressively invade local tissues and metastasize to lung, liver, and brain. Resection is rarely justified and 90% of patients are dead 12 months after diagnosis. Rhabdomyosarcomas are multicentric in 60% of patients, and are also bulky tumors with a strong tendency to invade local tissue and metastasize.

Secondary tumors

Secondary tumors are over 30 times more common than primary cardiac tumors. Approximately 1 in 10 malignant tumors eventually spread to heart or pericardium. The commonest malignancies that metastasize to heart or pericardium are leukemias, lymphomas, breast, lung, melanoma, and various sarcomas via blood and lymphatics. Direct spread from adjacent lung, breast, esophageal, and thymic tumors is common. Subdiaphragmatic tumors including uterine, renal, hepatic, and adrenal may invade the RA via the SVC. 10% of renal cell carcinomas invade the IVC and 40% of these reach the RA.

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